Desarrollo de una herramienta en Python para el postprocesado y comparación de múltiples casos transitorios pertenecientes a un modelo térmico diseñado en ESATAN-TMS

Trabajo de Fin de Grado

ESPECIALIDAD: Propulsión Aeroespacial

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Acronyms

1. Abstract

This Bachelor's Thesis focuses on the development of a Python-based post-processing tool for the thermal analysis of space systems using results generated with ESATAN-TMS.

Building upon a previous study limited to the post-processing of steady-state radiative cases, this work extends the analysis to fully transient thermal simulations. In addition, the tool has been designed to handle and compare multiple simulation cases associated with the same thermal model, allowing a more complete assessment of the system response under different operating conditions and orbital scenarios.

Since ESATAN-TMS presents limitations regarding integrated data visualization and case-to-case comparison, this project implements custom post-processing routines in Python, using libraries such as NumPy, Pandas, and Matplotlib. These routines automate the extraction, organization, and visualization of nodal results, including temperature evolution curves, transient heat flows, and comparative plots between different cases.

The resulting methodology improves the interpretation of complex transient thermal results and provides a more flexible framework for evaluating the thermal performance of a system throughout realistic orbital thermal cycles. By enabling the simultaneous analysis of several cases belonging to the same model, the developed tool supports a more efficient and systematic validation of thermal behaviour in space applications.

2. Introduction

Thermal management is an essential part of spacecraft design, as the temperature of each component must remain within allowable limits throughout the mission.

Since both the space environment and the operating conditions of the spacecraft evolve across different mission phases, steady-state cases cannot fully represent the real thermal behaviour. Transient simulations are therefore needed to examine the time-dependent behaviour of temperatures and heat flows.

The design of a spacecraft thermal control system relies on numerical analysis tools to predict system behaviour under different mission and operating conditions. These tools are used to represent the spacecraft through a thermal model and to calculate variables such as nodal temperatures and heat flows. Some of the best-known tools include THERMICA, Thermal Desktop together with SINDA/FLUINT, Simcenter 3D Space Systems Thermal (SST) and ESATAN-TMS, which provides a specialised environment for the development and solution of spacecraft thermal models.

The amount of information generated by these simulations can become considerable, particularly when transient analyses or multiple cases of the same model are considered. In such situations, the results must be organised and represented in a way that allows their evolution to be examined and the effects of the differences between cases to be identified. Dedicated post-processing tools can therefore support the interpretation of the simulation results.

This Bachelor's Thesis builds on an existing Python-based post-processing tool initially intended for steady-state analysis. The work combines the adaptation of the tool to transient and multi-case analysis with the development of a representative spacecraft thermal model in ESATAN-TMS. Selected thermal results are also used to perform a simplified electrical power analysis and provide the inputs for a separate battery model. The specific objectives of the project are presented in the following section.

2.1. Objectives

The main objective of the project is to enable the existing application to process, analyse, and compare several transient cases associated with the same ESATAN-TMS thermal model within a common environment.

Reaching this objective will require the coordinated development of the thermal model and the post-processing tool. The ESATAN-TMS model will provide a consistent set of transient results, while the application will be adapted to manage and compare them.

The thermal results will also be used to estimate the electrical power generated by the solar panels and to calculate the electrical power balance. These calculations will then provide the inputs for a separate simplified battery model.

Accordingly, the specific objectives of the project will be:

- Develop a representative spacecraft thermal model and define transient cases under different environmental and operating conditions.
- Adapt the existing application to load and manage several transient result files while retaining a common model hierarchy.
- Implement methods for representing and comparing temperatures, thermal loads, heat flows and exchanged thermal energy.
- Generate graphical outputs and summary tables, and allow the corresponding analyses and configurations to be saved, restored and exported.
- Estimate solar-panel electrical generation, calculate the electrical power balance and use the resulting quantities as inputs to a separate simplified battery model.
- Apply the complete analysis process to the selected transient cases and compare the thermal responses obtained under different environmental and operating conditions.

3. State of the Art

Spacecraft thermal analyses are commonly performed using specialised software tools, including THERMICA, Thermal Desktop (TD) coupled with SINDA/FLUINT, SST and ESATAN-TMS. [5–7]

The result files generated by ESATAN-TMS contain a considerable amount of data that must be organised, examined, and interpreted to evaluate the thermal behaviour of the model. Two thermal regimes are considered in the analysis. A steady-state analysis provides a time-independent thermal solution for a predefined set of boundary conditions, whereas a transient analysis describes the evolution of the calculated results throughout the simulated period [8, 9].

To facilitate the interpretation of these results, an interactive Python-based post-processing tool was developed as part of an earlier project [1]. The underlying Python code extracts and organises the results generated by ESATAN-TMS and presents them through numerical and graphical representations. The original implementation was tested primarily using steady-state results from HELIOS, a thermal model developed in ESATAN-TMS, so the original version focuses on the analysis of a single steady-state case. The present work extends the tool to process transient results and compare several cases associated with the same thermal model.

3.1. Post-Processing Capabilities in ESATAN-TMS

ESATAN-TMS supports the numerical solution of spacecraft thermal models under both steady-state and transient conditions. For each analysis case, the resulting dataset contains nodal temperatures, applied thermal loads, and conductive and radiative heat flows. The software provides numerical and graphical tools for examining these results [9, 10].

The structure of the exported information depends on the selected outputs and the type of analysis. Transient results associate each variable with successive output times, and heat-flow data identify the nodes or surfaces involved in each exchange. In every case, numerical identifiers must remain linked to the physical elements they represent [9, 10].

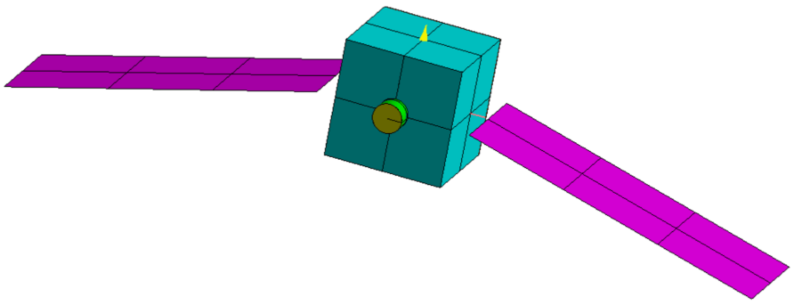
These datasets provide the information required to evaluate the thermal behaviour of the model. When transient analyses or several simulation cases are considered, however, direct inspection can become less practical. Relevant results must therefore be selected, organised, and compared consistently.

The post-processing functions available in ESATAN-TMS may not cover every requirement of a particular analysis. The Python implementation complements these functions by reorganising the exported thermal data according to an engineering hierarchy and providing additional methods for selecting, grouping, and representing the results [9].

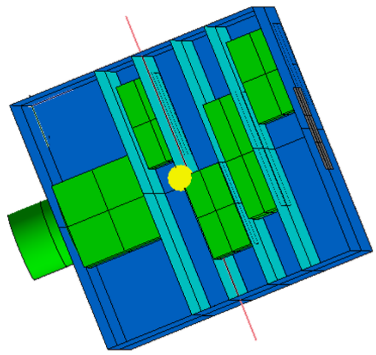
3.2. Steady-State Thermal Model: HELIOS

HELIOS is a thermal model developed in ESATAN-TMS with a single steady-state analysis case. The purpose of this model is to support the development and testing of the original Python-based tool. As a first approximation of a spacecraft thermal system, it provides sufficient detail to generate a consistent set of thermal results. The model contains the spacecraft structure, solar panels, batteries, radiating surfaces, and multi-layer insulation. The different components are connected through the conductive and radiative couplings that define the thermal network of the model [1].

The complete spacecraft configuration and a view of the internal components are shown in Figure 3.1.



(a) External view of the satellite model.



(b) Internal view of the satellite model.

Figure 3.1: External and internal views of the HELIOS thermal model. Taken from [1].

The steady-state solution of HELIOS contains the thermal data that are extracted and organised by the original Python code before being examined through the graphical interface. In contrast, the present work uses a separately developed spacecraft thermal model with several cyclic transient analysis cases.

3.3. Python-Based Post-Processing of Steady-State Cases

The original Python-based implementation uses the steady-state results generated for the HELIOS model in ESATAN-TMS [1]. In this type of analysis, a single temperature is calculated for each node under the selected boundary conditions, while the corresponding thermal loads and heat flows remain independent of time.

The imported results are linked to the hierarchy defined for the thermal model, so that each node can be located within the corresponding component and subsystem. Instead of working directly with the complete list of node identifiers contained in the ESATAN-TMS files, the user can examine the results through the spacecraft structure. This relationship between the thermal data and the physical model helps identify the origin of each value and interpret it in an engineering context.

The main modules of the original codebase and their connections are shown in Figure 3.2.

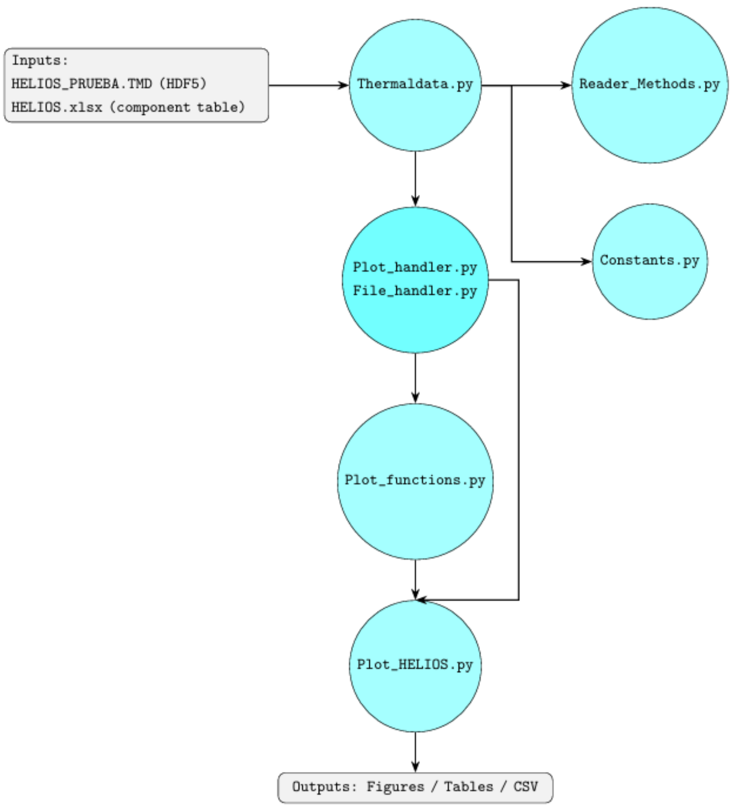


Figure 3.2: Post-processing architecture used in the previous project. Taken from [1].

The architecture separates result reading, data organisation and representation. `Thermaldata.py` imports the HELIOS result file and the external component table, while the supporting modules handle data selection, file management and plotting. The interface is provided by `Plot_HELIOS.py`, from which figures, tables and data files can be generated [1].

This version is suitable for examining the thermal results of an individual steady-state case. However, it does not provide complete post-processing of transient results or allow several simulation cases to be loaded and compared simultaneously. Extending the same structure to time-dependent results and multiple cases requires changes to both data management and representation.

3.4. Motivation for the Present Work

Transient simulations associate each calculated value with a specific simulation time. The post-processing tools must preserve this correspondence when results are selected, represented, and compared, particularly when several components or interfaces are considered together.

During an orbital cycle, transitions between solar illumination and eclipse, changes in internal dissipation, and the operation of active thermal-control elements produce variations in the thermal response of the spacecraft. These variations must therefore be examined throughout the orbital cycle to determine when they occur and which elements are most affected.

Comparing simulation cases introduces an additional requirement. Spacecraft thermal analyses may include several cases representing different environmental and operating conditions. Processing them separately requires equivalent elements to be selected again for every case and prevents their direct comparison within the same representation. Managing cases associated with the same thermal model together allows equivalent selections to be applied and the differences between them to be calculated consistently.

For these reasons, the existing implementation is extended to process transient results and manage several simulation cases associated with the same thermal model. This allows equivalent components and interfaces to be compared throughout the simulated period and the intervals in which the differences between cases are most significant to be identified.

4. Theoretical Background

Spacecraft temperatures are determined by the interaction between internal heat dissipation, external radiative loads and heat transfer through the structure. These contributions change throughout the mission, leading to different temperature conditions in each component. Their analysis requires an understanding of the space environment, the main heat-transfer mechanisms and the mathematical methods used to represent the physical system [2, 8, 11].

4.1. Spacecraft Thermal Control

Thermal control is required in many engineering systems because temperature affects equipment performance, material behaviour and reliability. Heat generated within a system and energy exchanged with the external environment must be managed so that the components remain within acceptable temperature limits [3].

In spacecraft, the Thermal Control System (TCS) maintains onboard components within their specified temperature limits throughout the mission, despite changes in environmental and operating conditions. Excessive temperatures, large thermal gradients and repeated thermal cycles can shorten component life, reduce measurement accuracy or prevent equipment from functioning correctly. The design must also satisfy the restrictions imposed by the available mass, volume and electrical power [2, 11, 12].

Thermal requirements depend on the function and operating state of each component. They include allowable temperature ranges, limits on thermal gradients and requirements for temperature stability. For example, electrical and electronic equipment, including batteries and onboard computers, operate within specified temperature ranges. Optical instruments such as telescopes often require tighter control of temperature gradients to limit dimensional changes and maintain alignment [11, 12].

The temperature of each component depends on the balance between internally dissipated power and the energy exchanged with the external environment. This balance is influenced by the amount of energy absorbed by the external surfaces, the heat transferred through the internal structure and the energy emitted towards space. Since these contributions change during the mission, the required control must remain effective under different operating and environmental conditions [2, 11].

Different thermal-control measures are combined to meet these requirements. Surface coatings modify solar absorption and infrared emission, radiators provide suitable surfaces for heat rejection, and Multi-Layer Insulation (MLI) limits unwanted radiative exchange. Conductive links distribute heat between components and structural elements, while heaters provide additional energy when passive measures are insufficient under cold conditions [2, 11, 13].

The selection of these measures depends on the permitted temperature ranges, expected heat loads and mission phases. Thermal control is therefore not limited to maintaining a single uniform temperature. It must also control gradients, reduce excessive variations and meet the individual requirements of each component [11, 12].

4.2. Space Thermal Environment

The space environment is one of the main factors affecting heat exchange through the external surfaces of a spacecraft. In orbit, energy is exchanged with space mainly by radiation because external convective transport is absent. Conduction remains important within the structure and between connected components, but it does not provide a direct mechanism for rejecting heat to space [2, 11].

For a spacecraft orbiting the Earth, the main environmental heat inputs are direct solar radiation, solar radiation reflected by the Earth, known as albedo, and infrared radiation emitted by the planet. At the same time, the external surfaces emit thermal radiation towards deep space. These exchanges are represented in Figure 4.1 [2, 11, 13].

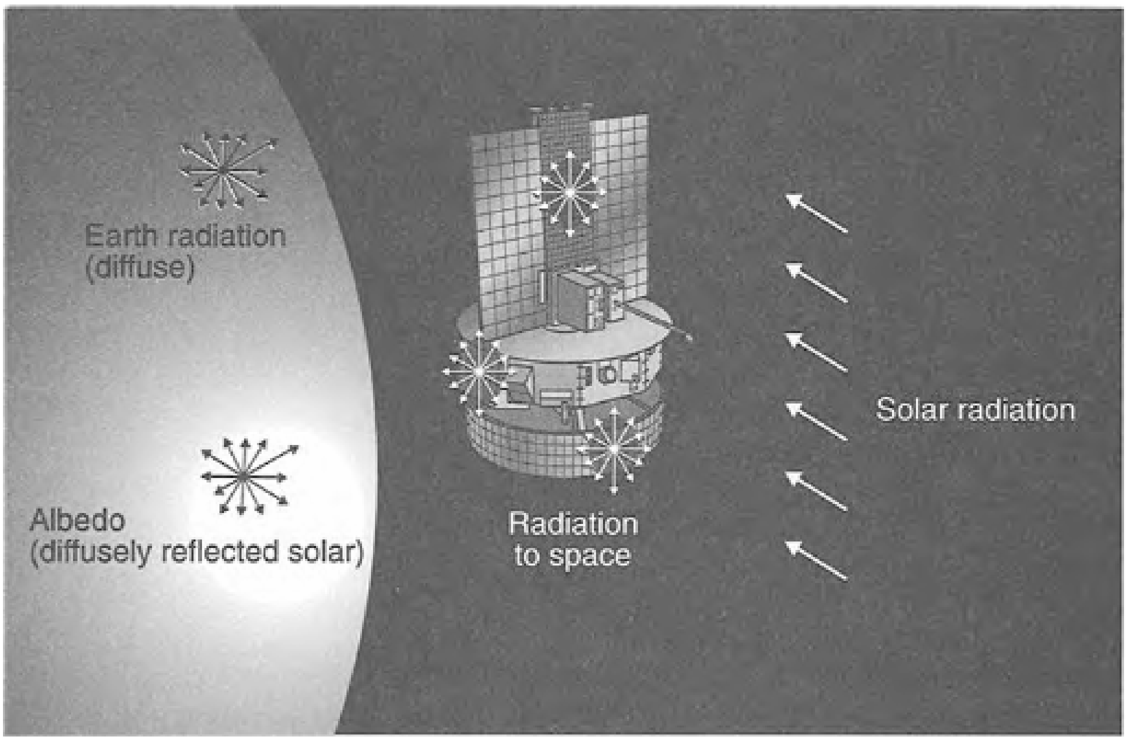


Figure 4.1: Main radiative exchanges between an orbiting spacecraft and the external environment. Taken from [2].

Direct solar radiation generally provides the largest environmental heat input for an Earth-orbiting spacecraft. The power absorbed by a flat surface depends on the solar irradiance, surface area, orientation and solar absorptivity. It can be expressed as [11]:

$$\dot{Q}_{\mathrm{sun}} = \alpha_{\mathrm{s}} G_{\mathrm{s}} A \cos\theta \tag{4.1}$$

where α_{s} is the solar absorptivity, G_{s} is the solar irradiance, A is the surface area and θ is the angle between the surface normal and the solar rays. Since G_{s} is expressed in $\mathrm{W\,m^{-2}}$, the resulting absorbed power is expressed in W [11].

Albedo is solar radiation reflected by the Earth's surface and atmosphere. The albedo's contribution depends on the illuminated portion of the planet visible from the spacecraft and on the fraction of reflected energy absorbed by each exposed surface. Planetary infrared radiation instead originates from the Earth's thermal emission and depends on the infrared properties of the exposed surfaces. In both cases, the incoming energy also depends on the portion of the planet visible from each surface [2, 11].

The contribution made by each source varies with the orbit, spacecraft attitude, exposed area and thermo-optical properties. During a total eclipse, the direct-solar term becomes zero, and rapid variations occur when the spacecraft enters or exits the eclipse region. Albedo follows the visible illuminated region of the Earth, whereas planetary infrared radiation remains present during eclipse [11, 13].

The periodic sequence of illuminated and eclipse phases exposes the spacecraft to alternating heating and cooling during each orbit. Component temperatures adjust over a finite interval because energy is stored and transferred through the structure. A time-dependent analysis is required to represent this delayed response to the changing environmental loads [2, 11].

4.3. Heat Transfer and Thermal Energy Storage

The temperature response of a spacecraft depends on the amount of heat transferred and stored over time. Heat transfer occurs by conduction, convection or radiation. Conduction and radiation are the main mechanisms in spacecraft thermal analysis, while thermal capacitance accounts for the energy stored by each component [3, 11].

4.3.1. Conduction Heat Transfer

Conduction transfers thermal energy through a material or between bodies in contact without bulk movement of matter. The conductive heat flow depends on the temperature gradient, the thermal conductivity of the material and the area normal to the direction of transfer. For one-dimensional conduction, Fourier's law is written as [3, 4]

$$\dot{Q}_x = -kA\frac{\mathrm{d}T}{\mathrm{d}x}, \tag{4.2}$$

where $\dot{Q}_x$ is the conductive heat flow, k is the thermal conductivity, A is the area normal to the heat flow and $\mathrm{d}T/\mathrm{d}x$ is the temperature gradient. The negative sign indicates that heat flows in the direction of decreasing temperature.
Figure 4.2 shows the one-dimensional plane-wall configuration described by Fourier's law.

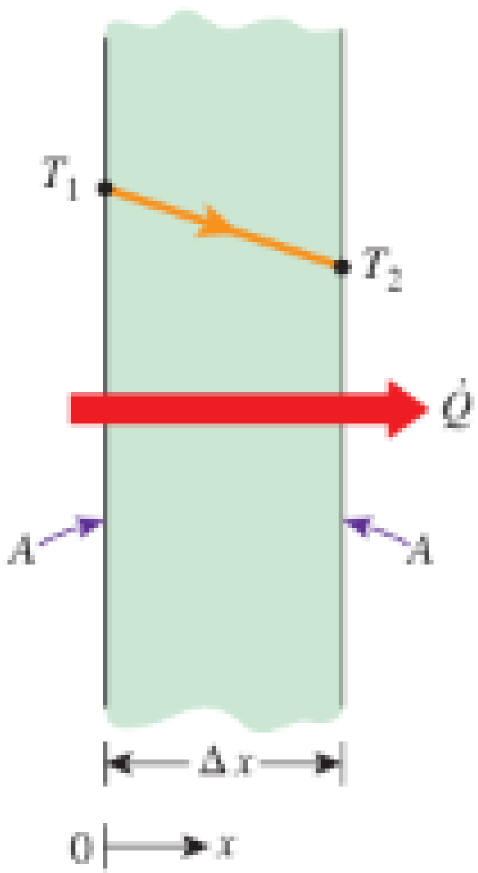


Figure 4.2: One-dimensional heat conduction through a plane wall. Taken from [3].

For a plane layer of thickness L, constant area and constant conductivity, Fourier's law can also be expressed in terms of thermal resistance [3, 4]:

$$R_{\text{cond}} = \frac{L}{kA} \tag{4.3}$$

$$\dot{Q}_{12} = \frac{T_1 - T_2}{R_{\text{cond}}} \tag{4.4}$$

The thermal-resistance formulation allows conductive paths to be combined in series or parallel. At an interface between two components, the effective resistance also depends on surface roughness, contact pressure and any intermediate materials. These effects are represented by the contact resistance, while the linear thermal conductance is defined as the inverse of that resistance [2,3]:

$$G^{L}_{12} = \frac{1}{R_{12}} \tag{4.5}$$

where G^{L}_{12} is the linear thermal conductance between components 1 and 2, and R_{12} is the corresponding resistance. This relation is particularly useful in nodal models because each conductive connection is represented by a conductance between two nodes. When temperature changes with position or time, Fourier's law is combined with conservation of energy to obtain the heat-diffusion equation [3,4,8].

4.3.2. Radiation Heat Transfer

Unlike conduction, thermal radiation transfers energy through electromagnetic waves, which propagate through vacuum. Radiative exchange takes place between spacecraft surfaces, while external surfaces also emit energy towards space. The power emitted by a real surface depends on the absolute temperature, area and emissivity, as expressed by the Stefan–Boltzmann law [3,11]:

$$\dot{Q} = \varepsilon \sigma A T^4 \tag{4.6}$$

where ε is the surface emissivity, σ is the Stefan–Boltzmann constant, A is the emitting area and T is the absolute surface temperature. The value of the Stefan–Boltzmann constant is
$\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$ [3].
For an external surface facing deep space without obstruction, the net radiative heat flow is expressed as [11]:

$$\dot{Q}_{\text{rad}} = \varepsilon \sigma A \left(T^4 - T^4_{\text{space}}\right) \tag{4.7}$$

where T_{space} is the effective temperature assigned to deep space. The term T^4_{space} is generally small because the deep-space temperature is much lower than the spacecraft surface temperature [11].

The Stefan–Boltzmann law gives the total power emitted over all wavelengths. The spectral distribution, however, depends on temperature. As temperature increases, the emitted power rises and the maximum moves towards shorter wavelengths. This behaviour explains why radiation received from the Sun is mainly treated as short-wave radiation, whereas the Earth and spacecraft surfaces emit primarily within the infrared range, as illustrated in Figure 4.3 [3, 11].

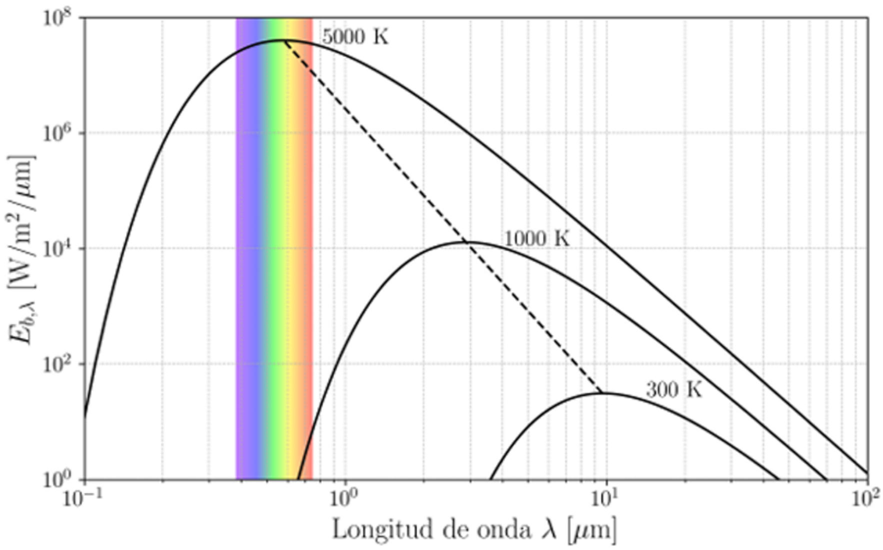


Figure 4.3: Blackbody spectral emissive power at different temperatures and displacement of the emission maximum. Taken from [4].

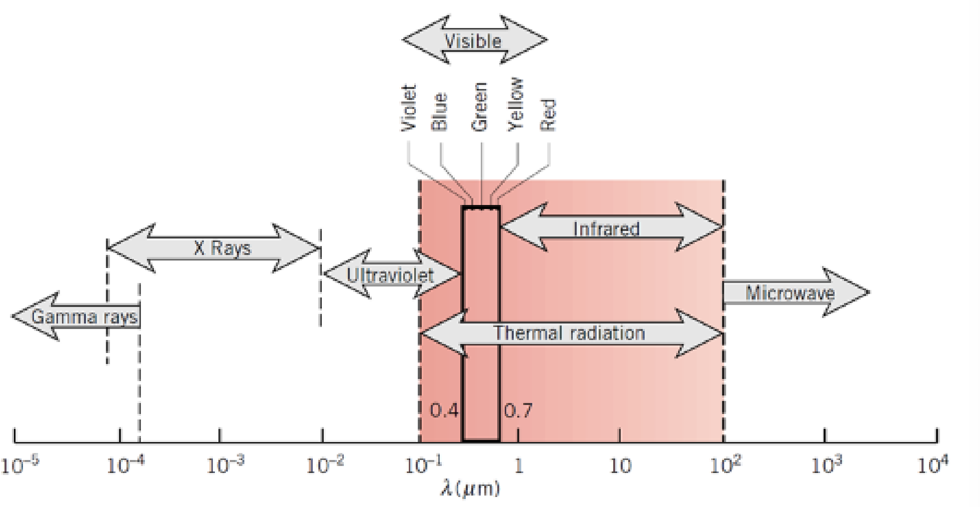
When radiation reaches a surface, part of it is absorbed, part is reflected and the remainder, if any, is transmitted. Conservation of radiative energy requires the corresponding fractions to satisfy [3]

$$\alpha + \rho + \tau = 1 \tag{4.8}$$

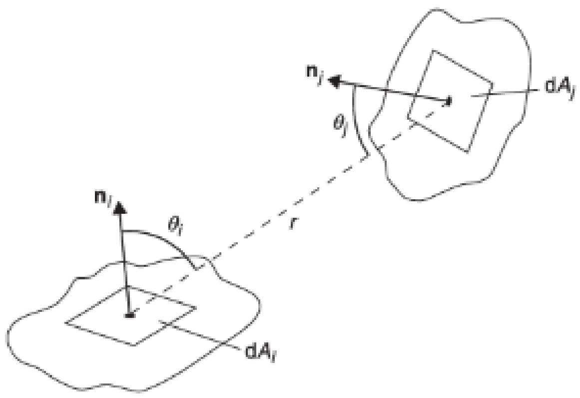
where α, ρ and τ are absorptivity, reflectivity and transmissivity, respectively. For an opaque surface, no radiation is transmitted, and Equation 4.8 reduces to $\alpha + \rho = 1$ [3, 11].

Since these properties depend on wavelength, the solar and infrared bands are treated separately in spacecraft thermal analysis. Solar absorptivity represents the fraction of short-wave solar radiation absorbed by a surface, while infrared emissivity governs the exchange of long-wave thermal radiation. The same surface can therefore have different properties in the two bands [11,13,14]. The wavelength range associated with thermal radiation is shown in Figure 4.4a.

The view factor F_{ij} represents the fraction of radiation leaving surface i that reaches surface j. The value of F_{ij} depends on the relative position, orientation and area of the two surfaces, as illustrated in Figure 4.4b [11].



(a) Wavelength range of thermal radiation. Taken from [14].



(b) Geometry used to define a view factor. Taken from [11].

Figure 4.4: Wavelength dependence and geometrical definition of thermal radiation.

View factors satisfy the reciprocity relation. When a set of N surfaces forms a closed enclosure, the factors from each surface also satisfy the following rule [3, 11]:

$$A_i F_{ij} = A_j F_{ji} \tag{4.9}$$

$$\sum_{j=1}^{N} F_{ij} = 1 \tag{4.10}$$

where N is the number of surfaces forming the enclosure.

Radiative exchange depends on the surface temperatures and thermo-optical properties, as well as on the areas and relative geometry of the surfaces. Analytical view factors are available for simple arrangements, whereas spacecraft radiative models generally calculate them numerically [11,13].

4.3.3. Thermal Capacitance and Transient Response

Thermal capacitance measures the energy required to change the temperature of a body. For a body with uniform temperature and constant properties, it is given by [2,3]:

$$C = mc_p \tag{4.11}$$

$$\Delta E = C\Delta T \tag{4.12}$$

where C is the thermal capacitance, m is the mass, c_p is the specific heat capacity and ΔE represents the changes in the stored thermal energy of a body.

For a transient process, the energy balance is expressed as [11]:

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = \dot{Q}_{\mathrm{net}} \tag{4.13}$$

where $\dot{Q}_{\mathrm{net}}$ is the net heat flow into the body. For the same net heat flow, a larger thermal capacitance results in a slower temperature change [3].

For a single body connected to a constant-temperature boundary through a linear thermal resistance R, the thermal time constant and the corresponding temperature response are expressed as [3]:

$$\tau_{\mathrm{th}} = RC \tag{4.14}$$

$$T(t) = T_\infty + (T_0 - T_\infty)\exp\left(-\frac{t}{\tau_{\mathrm{th}}}\right) \tag{4.15}$$

where T_0 is the initial temperature and T_∞ is the equilibrium temperature approached as time increases. A larger time constant corresponds to a slower response [3].

During a transient process, heat flow changes with time, and the net energy transferred between two instants is obtained by integrating the heat flow over the corresponding interval [3]:

$$E_{12} = \int_{t_1}^{t_2} \dot{Q}(t)\,\mathrm{d}t \tag{4.16}$$

The sign of E_{12} follows the sign convention adopted for $\dot{Q}$. In orbital analysis, this integral quantifies the energy absorbed or released during a selected part of the orbit. For example, energy is stored during an illuminated interval and released after eclipse entry. The corresponding temperature change is delayed by the thermal capacitance by the thermal capacitance and the conductive and radiative connections of the component with the rest of the system [2, 11].

4.4. Mathematical Modelling of Spacecraft Thermal Systems

Spacecraft thermal models combine the equations governing heat transfer and energy storage. The principle of conservation of energy is applied to calculate temperatures and heat flows under specified loads and boundary conditions. Analytical methods provide exact solutions when the geometry and boundary conditions are simple. More complex configurations, involving multiple materials, interfaces and time-dependent inputs, are formulated as a finite system of equations for numerical analysis [11, 13].

4.4.1. Spatial Discretisation Approaches

The numerical solution of heat-transfer equations requires a discrete representation of the geometry and temperature field. The geometry is divided into a finite number of regions, each associated with one or more temperature values. This process is known as spatial discretisation [11, 13].

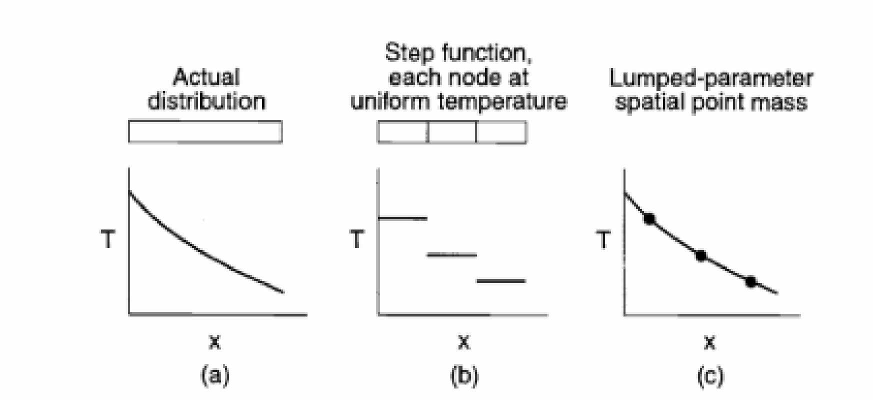
Several numerical methods are available to discretise the thermal problem. Finite-difference methods express spatial derivatives through temperature differences between adjacent grid points. The Finite-Element Method (FEM) divides the geometry into connected elements and describes the temperature variation within each element using interpolation functions. Finite-volume methods apply an energy balance to each control volume using the heat flows across the control-volume boundaries. The three formulations differ, but all of them reduce the thermal problem to a finite system of equations expressed in terms of discrete temperatures [2, 13].

At system level, spacecraft thermal models commonly use a network of lumped nodes. Each node represents a particular region of the model with a single temperature value. Regions where small temperature gradients are expected can be represented by one node, whereas regions with larger gradients or relevant heat-transfer paths are divided into several nodes [11, 13].

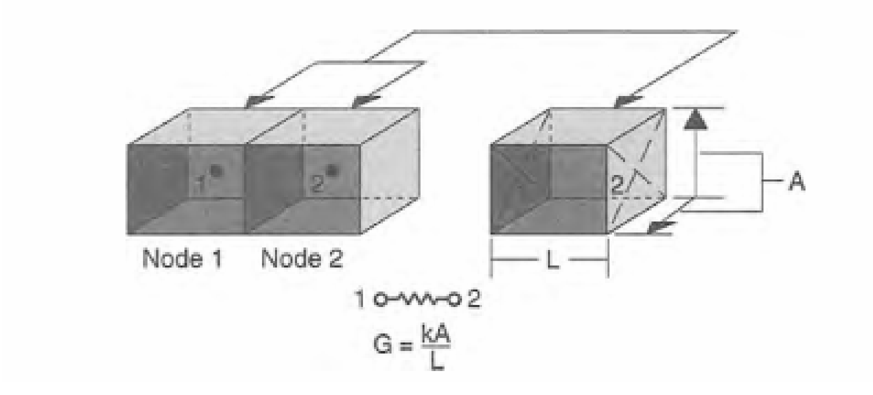
A finer division provides greater spatial resolution, but it also increases the number of conductive and radiative couplings, the calculation time and the volume of output data. The selected nodalisation must resolve the temperatures, gradients and heat paths relevant to the analysis without introducing unnecessary detail [2, 13].

4.4.2. Lumped Parameter Method (LPM)

In the LPM, a thermal system is represented as a network of discrete nodes linked by conductive and radiative couplings. Each node represents a region whose temperature is assumed to be spatially uniform, while the energy stored in that region is represented by a thermal capacitance. Figure 4.5 illustrates the approximation of a continuous temperature distribution by discrete nodal values and the conductive connection between two adjacent nodes [2, 8].



(a) Approximation of a continuous temperature distribution by discrete nodal values.



(b) Conductive connection between two adjacent nodes.

Figure 4.5: Temperature approximation and conductive coupling in a lumped-parameter model. Taken from [2].

The principle of conservation of energy is applied to each of the N capacitive nodes to establish the corresponding energy balance. This converts the continuous temperature field into a coupled system of N first-order ordinary differential equations, one for each nodal temperature. The corresponding balance for a general node i is written as [8, 11]:

$$C_i \frac{\mathrm{d}T_i}{\mathrm{d}t} = \dot{Q}_i(t) + \sum_{\substack{j=1 \\ j\neq i}}^{N} G_{ij}^{L} \left(T_j - T_i\right) + \sigma \sum_{\substack{j=1 \\ j\neq i}}^{N} G_{ij}^{R} \left(T_j^4 - T_i^4\right) \tag{4.17}$$

where C_i is the nodal thermal capacitance, $\dot{Q}_i(t)$ is the net heat load applied to the node, G_{ij}^{L} is the linear conductance, G_{ij}^{R} is the radiative exchange coefficient, σ is the Stefan–Boltzmann constant and N is the total number of nodes in the network.

Conductive exchange is proportional to the temperature difference $T_j - T_i$, whereas radiative exchange depends on $T_j^4 - T_i^4$. This T^4 dependence makes the resulting system non-linear [8, 11].

The nodal equations are completed by assigning the applied loads and boundary conditions. The loads represent internally dissipated power, external radiative heating or heat supplied by temperature-controlled devices. Boundary conditions specify temperatures, heat inputs or other interactions that are not calculated as unknown nodal temperatures [8, 10].

A transient calculation also requires an initial temperature for each capacitive node. This initial distribution affects the beginning of the solution and is obtained from an estimated condition, a previous transient interval or a steady-state calculation. The influence of this initial distribution must be considered when the required result is a periodic orbital response [8, 10].

4.4.3. Steady-State and Transient Formulations

In a steady-state solution, the nodal temperatures remain constant. As a result, $\mathrm{d}T_i/\mathrm{d}t$ is zero, and the storage term in Equation (4.17) disappears. The resulting non-linear algebraic equations determine the steady-state temperatures for fixed loads and boundary conditions. However, they contain no information about the transition towards this equilibrium state [8, 11].

In a transient solution, the storage term is retained and an initial temperature is assigned to each capacitive node. The coupled differential equations are integrated over time, allowing the loads and boundary conditions to change during the selected interval. The resulting solution describes temperatures and heat flows as functions of time and includes the energy stored in the capacitive nodes [8, 11].

A transient calculation can therefore identify temporary maximum and minimum temperatures, the time at which they occur and the delay between a change in load and the corresponding temperature response. The selected formulation depends on whether only an equilibrium condition is required or whether the evolution of the system must also be represented [2, 13].

4.4.4. Periodic Thermal Response and Cyclic Convergence

When the orbital geometry and operating sequence repeat after a period P, the applied loads are also periodic. The thermal response is periodic when the nodal temperatures repeat after the same period. To approach this condition, the final temperature distribution from each calculated cycle is used to initialise the next one [10,11,15].

$$\dot{Q}_i(t+P) = \dot{Q}_i(t) \tag{4.18}$$

$$T_i(t+P) = T_i(t) \tag{4.19}$$

In numerical calculations, small differences generally remain between the temperatures calculated at the beginning and end of a complete period. Cyclic convergence is accepted when the maximum of these differences is below a selected tolerance [8,10].

$$\text{máx}\,|T_i(t_0+P) - T_i(t_0)| \leq \varepsilon_{\text{cyc}} \tag{4.20}$$

where ε_{cyc} is the convergence criterion. Once this condition is satisfied, the calculated cycle represents the response to the periodic loads rather than the initial numerical adjustment of the model [8,10,15].

4.5. Thermal Modelling and Solution Framework in ESATAN-TMS

ESATAN-TMS applies a lumped-parameter nodal formulation through connected geometrical, radiative and thermal models. The physical description is converted into a network containing thermal capacitances, conductive and radiative couplings, boundary conditions and applied loads [8,9,16].

The modelling process links the physical configuration with the equations solved during the analysis. Geometrical and surface information is used in the radiative calculations, while material properties and physical connections determine thermal storage and conduction. These contributions are combined in the thermal network before the selected numerical solution is performed [8,9].

4.5.1. Geometrical and Radiative Models

The geometrical model defines the dimensions, positions and orientations of the modelled elements, together with their materials and thermo-optical properties. The geometrical discretisation also establishes the regions that can later be associated with individual thermal nodes [9,16].

The corresponding surfaces form the radiative model. View factors and radiative exchange factors between surfaces are calculated from their geometry and properties. During an orbital analysis, the model also determines the solar, albedo and planetary infrared loads absorbed at the selected orbital positions [9,13,16].

4.5.2. Thermal Network Representation

The thermal network consists of capacitive nodes linked by conductors. Each capacitive node represents a region with an associated temperature and heat-storage capacity. The nodal capacitance is calculated from the mass and specific heat assigned to the corresponding part of the model [8, 9].

Linear conductors represent solid conduction, contact conductance and other couplings proportional to temperature difference. Radiative conductors represent energy exchange between surfaces and are obtained from the radiative calculation. Environmental inputs, internal dissipation and boundary conditions complete the nodal energy balances [8–10].

Nodes and conductors are generated from the geometry or defined explicitly when a physical element or connection is not represented directly by the geometrical model. This allows equipment units, dedicated conductive links and user-defined thermal interactions to be included without adding unnecessary geometrical detail [8, 9].

4.5.3. Implementation of Steady-State, Transient and Cyclic Solutions

The steady-state solver determines the nodal temperatures for fixed loads and boundary conditions. The transient solver retains the nodal capacitances and initial temperatures and integrates the energy-balance equations over time. Calculation and output times are selected according to the temporal resolution required for the analysis [8, 10].

For periodic cases, the cyclic solver repeats the defined interval and uses the final temperature distribution of one cycle to initialise the next. The process continues until the difference between consecutive cycles satisfies the selected convergence criterion or the permitted number of cycles is reached [8, 10, 16].

The numerical solution provides the nodal temperatures and the contributions included in their energy balances. Depending on the selected outputs, these results may include applied loads and conductive and radiative heat flows, which can subsequently be examined as steady values or as functions of time [9, 10].

5. Thermal Model Development in ESATAN-TMS and Transient Analysis Cases

The thermal model developed in ESATAN-TMS generates the results analysed throughout this work. It does not reproduce a particular flight design in full detail. Instead, it represents the main heat-transfer interactions with sufficient geometrical and physical detail to generate transient results and compare different cases. The same geometry, node hierarchy and thermal network are retained in every simulation, while the environmental and operating conditions are modified between cases.

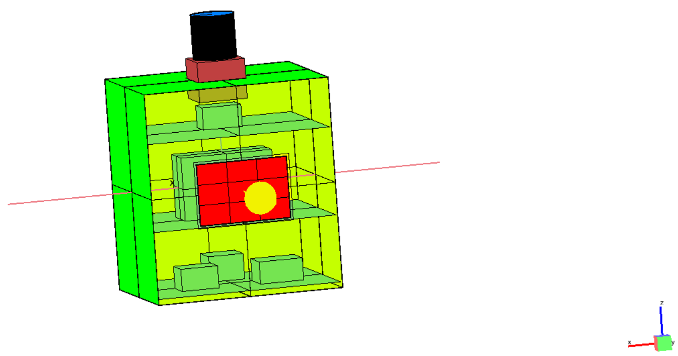
The model contains the main body, two solar-panel wings, a camera, several internal electronic units, a battery, a radiator and Multi-Layer Insulation (MLI). Conductive and radiative couplings connect these elements, allowing temperatures and heat flows to be calculated throughout each simulation. The model follows the lumped-parameter formulation presented in Chapter 4 and implemented in ESATAN-TMS [8, 9].

5.1. Model Development and Geometrical Configuration

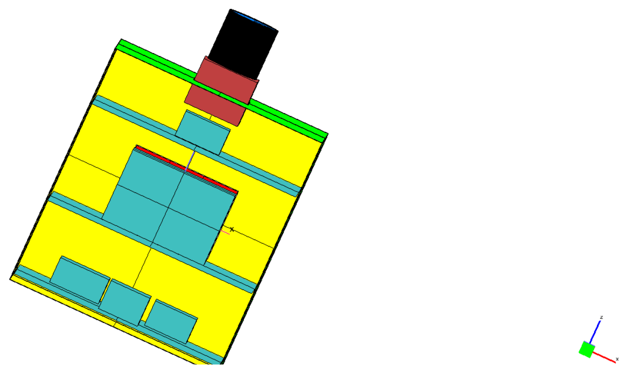
The geometrical model is refined to represent the arrangement of the internal plates and equipment and the main conductive connections between them. This organisation also maintains a direct link between the physical elements and the node hierarchy used by the post-processing application.

The final internal configuration is shown from the $+Y$ and $-Y$ sides in Figure 5.1. Both views show the distribution of the equipment across the structural plates and the different levels of the main body. The dissipative components represented in the model are the camera electronics, the battery, the On-Board Computer (OBC), the communications unit and the Attitude Determination and Control System (ADCS). Each unit is represented separately, allowing an individual internal heat load to be assigned and the corresponding temperature and heat exchanges to be analysed independently

The final external configuration is shown in Figure 5.2. The main body presents a rectangular geometry and is divided into several surface nodes. Two solar-panel wings extend from opposite sides and are discretised independently. The camera assembly extends from the main body. Heat is rejected to space through the radiator, while the MLI-covered surfaces limit heat exchange with the external environment.



(a) Internal configuration viewed from the $+Y$ side.



(b) Internal configuration viewed from the $-Y$ side.

Figure 5.1: Internal configuration of the spacecraft thermal model viewed from opposite sides. The colours correspond to the geometry layer and do not represent temperatures.

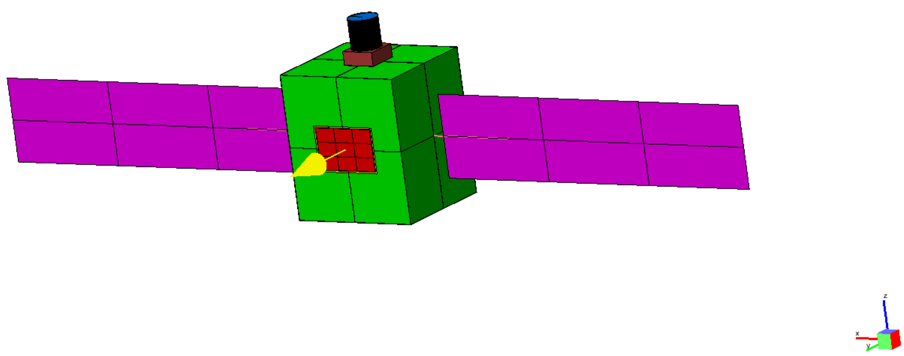


Figure 5.2: Final external configuration of the spacecraft thermal model in ESATAN-TMS. The colours correspond to the geometry layer and do not represent temperatures.

5.2. Thermal Model Hierarchy

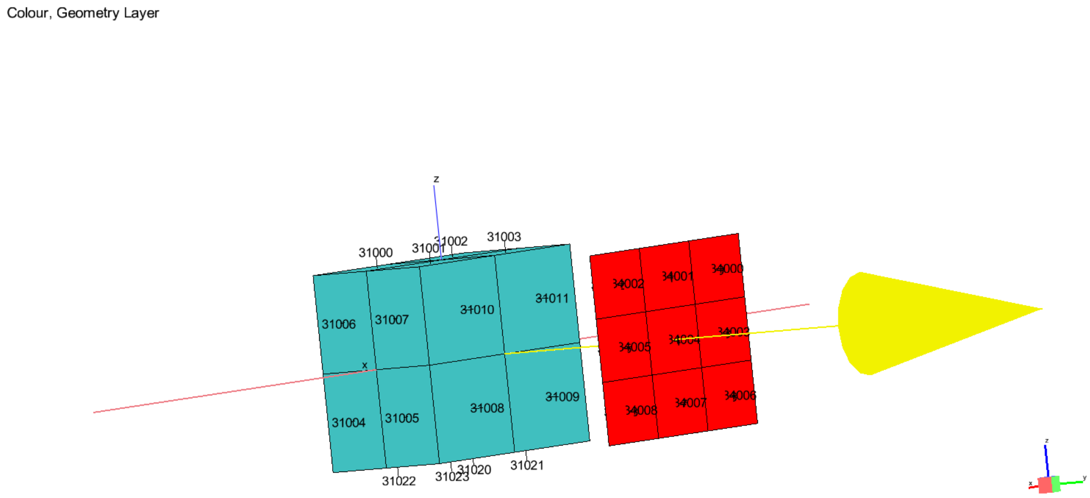
The numerical nodes are grouped according to the physical elements they represent. This hierarchy consists of four levels:

- **Level 0:** includes the complete `SAT_AFM` model and the `SPACE` boundary.
- **Level 1:** separates the main structure from the solar-panel assembly.
- **Level 2:** divides the structure into the body, camera, avionics and MLI groups.
- **Level 3:** contains the individual panels, structural plates, camera elements and equipment units.

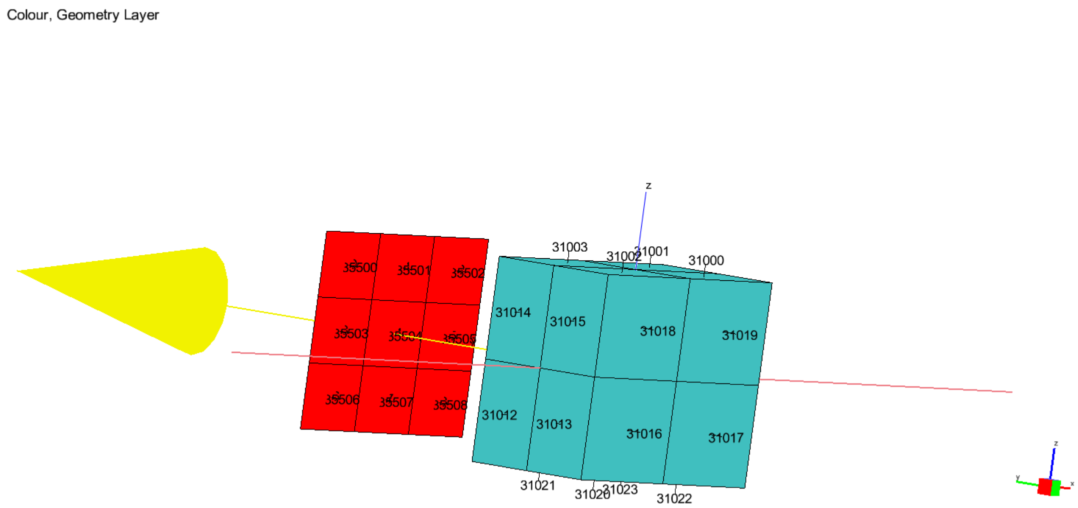
Each group is identified by a node range and a short label, giving a consistent naming system throughout the model. The *Power* field indicates the equipment groups which have an internal heat load applied. The complete classification is given in Table 5.1.

Table 5.1: Thermal-model hierarchy and node identifiers.

Group	**Level**	**Parent**	**ID**	**Label**	**Power**
Space	0	None	#99999	SPACE	False
SAT_AFM	0	None	#10000–75250	SAT_AFM	False
STRUCTURE	1	SAT_AFM	#10000–51150	STR	False
SOLAR_PANELS	1	SAT_AFM	#70000–75250	SP	False
BODY	2	STRUCTURE	#10000–13650	STR_BOD	False
CAMERA	2	STRUCTURE	#40000–43750	STR_CAM	False
AVIONICS	2	STRUCTURE	#30000–35550	STR_AV	False
MLI	2	STRUCTURE	#50000–51150	STR_MLI	False
SOLAR_PANEL_1	3	SOLAR_PANELS	#70000–72550	SP_1	False
SOLAR_PANEL_2	3	SOLAR_PANELS	#73000–75250	SP_2	False
BODY_PANEL	3	BODY	#10000–11150	STR_BOD_P	False
BODY_PLATE_INF	3	BODY	#12000–12550	STR_BOD_PLI	False
BODY_PLATE_MED	3	BODY	#12600–13050	STR_BOD_PLM	False
BODY_PLATE_SUP	3	BODY	#13100–13650	STR_BOD_PLS	False
CAMERA_BASE	3	CAMERA	#40000–40450	STR_CAM_BASE	False
CAMERA_ELEC	3	CAMERA	#40500–41050	STR_CAM_ELEC	True
CAMERA_LENS	3	CAMERA	#41500–42150	STR_CAM_LENS	False
CAMERA_CYL	3	CAMERA	#42500–43550	STR_CAM_CYL	False
BATTERY	3	AVIONICS	#30000–30550	STR_AV_BAT	True
OBC_BOX	3	AVIONICS	#31000–31550	STR_AV_OBC	True
COMMS_BOX	3	AVIONICS	#32000–32550	STR_AV_COM	True
ADCS_UNIT	3	AVIONICS	#33000–33550	STR_AV_ADCS	True
RADIATOR	3	AVIONICS	#34000–35550	STR_AV_RAD	False
MLI_PANELS	3	MLI	#50000–51150	STR_MLI_P	False



(a) First view of the OBC and radiator nodal numbering.



(b) Opposite view of the OBC and radiator nodal numbering.

Figure 5.3: Example of nodal numbering for the OBC and adjacent radiator surfaces, shown from opposite sides. The figure illustrates the node ranges listed in Table 5.1.

5.3. Material and Surface Properties

The temperature response of each geometrical element depends on the assigned bulk material and the thermo-optical properties of the surfaces. Thermal conductivity, density and specific heat capacity determine heat conduction and energy storage, while the surface properties govern the interaction with solar and infrared radiation. The same definitions are used in all analysis cases.

Consequently, changes in the calculated results are associated with the environmental and operating inputs rather than with modifications to the model.

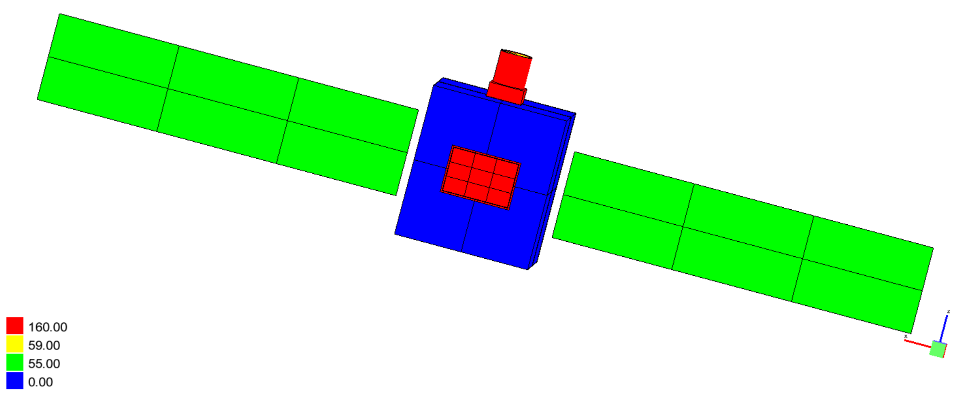
5.3.1. Bulk Properties

The geometrical elements use four material definitions. Aluminium 6061 is assigned to the main structural panels, internal plates and equipment housings. Gallium arsenide represents the solar-cell elements, while a separate optical material is used for the camera lens. The MLI foil has an independent definition whose conductivity is set to zero. This prevents the geometrical shell from introducing a solid-conduction connection, while the intended couplings are defined explicitly in the thermal network.

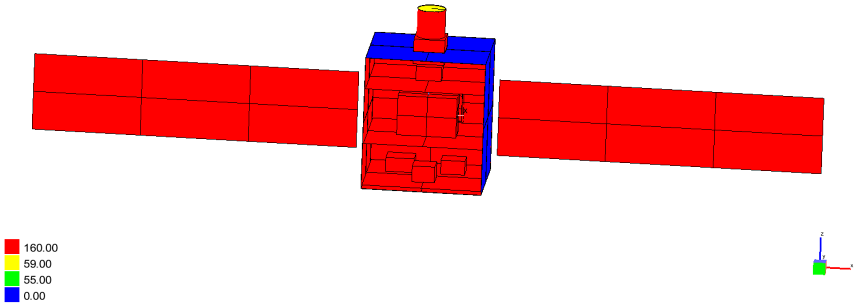
The conductivity, specific heat capacity and density assigned to each material are listed in Table 5.2. Figure 5.4 shows only the conductivity distribution, using two opposite views to cover the complete geometry. This allows the assignments to the structure, solar cells, camera and MLI surfaces to be checked.

Table 5.2: Material properties used in the thermal model.

Name	$\boldsymbol{k}$ [$\mathrm{W\,m^{-1}\,K^{-1}}$]	$\boldsymbol{c_p}$ [$\mathrm{J\,kg^{-1}\,K^{-1}}$]	$\boldsymbol{\rho}$ [$\mathrm{kg\,m^{-3}}$]
Al_6061	160	900	2700
GaAs	55	1000	5300
Glass	59	310	5330
MLI_foil	0	900	300



(a) View from the side $+Y$.



(b) View from side $-Y$.

Figure 5.4: Thermal conductivity distribution over the final spacecraft model, viewed from opposite sides.

5.3.2. Thermo-optical Properties

The surface definitions are assigned according to the function of each element. Anodised aluminium is used on selected structural surfaces, while black paint is applied to the internal equipment. An Optical Solar Reflector (OSR) defines the radiator surface, and the solar cells use a separate coating definition. The external MLI is represented by Kapton coated with Indium Tin Oxide (ITO), while independent properties are assigned to the internal MLI and the infrared lens.

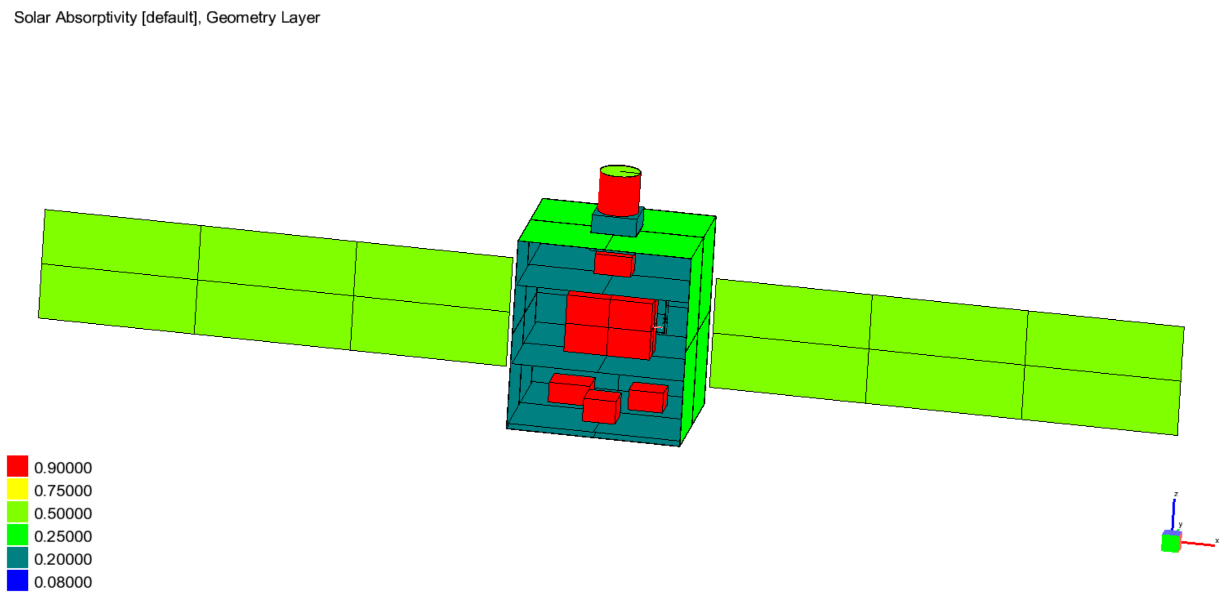
The solar and infrared bands are defined separately. For each surface, the model specifies absorptivity or emissivity, transmissivity and the specular and diffuse components of reflectivity. Within each band, these coefficients are defined so that the corresponding energy fractions sum to unity.

The adopted solar absorptivity and infrared emissivity values are consistent with representative measurements compiled for black coatings, anodised aluminium, solar cells, optical solar reflectors and Kapton films [17]. These data are used as reference values, since a single nominal definition is assigned to each surface in the model. The complete set of properties is summarised in Table 5.3.

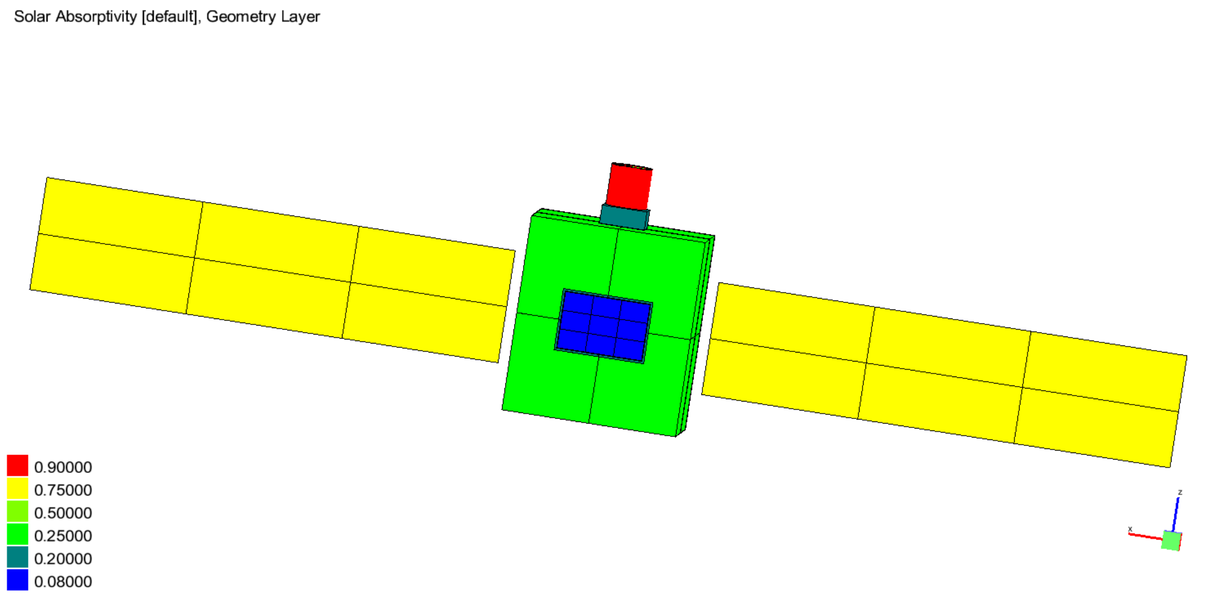
Table 5.3: Thermo-optical coating properties used in the thermal model.

Name	**Infrared**				**Solar**			
	ε	τ	ρ_{spec}	ρ_{diff}	α	τ	ρ_{spec}	ρ_{diff}
Al anodized	0.6	0	0	0.4	0.2	0	0	0.8
Black paint	0.9	0	0	0.1	0.9	0	0	0.1
OSR	0.8	0	0.2	0	0.08	0	0.92	0
Solar cells	0.75	0	0	0.25	0.75	0	0	0.25
IR lens	0.1	0.9	0	0	0.5	0	0	0.5
Kapton ITO	0.8	0	0	0.2	0.5	0	0	0.5
MLI_int	0.2	0	0	0.8	0.2	0	0	0.8

The solar absorptivity and reflectivity distributions are presented in Figures 5.5 and 5.6. Higher absorptivity is assigned to the solar cells and black-painted surfaces, whereas the radiator OSR has the lowest value. The reflectivity maps show the corresponding distribution of reflected solar energy. Two opposite views are included for each property so that the assignments can be checked over the complete geometry.

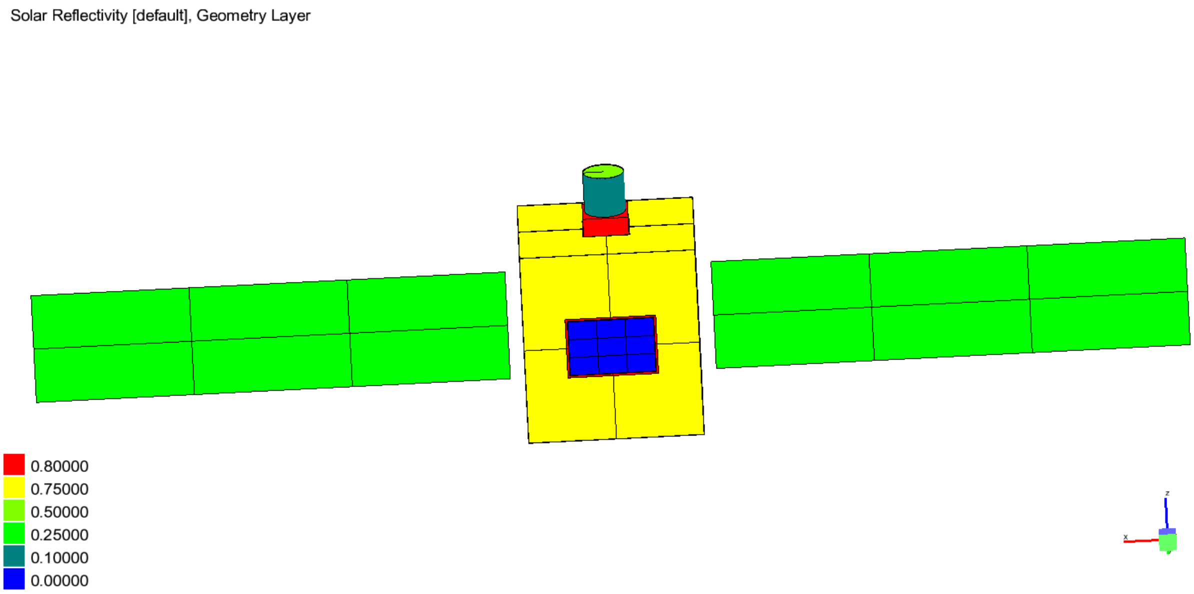


(a) View from one side.

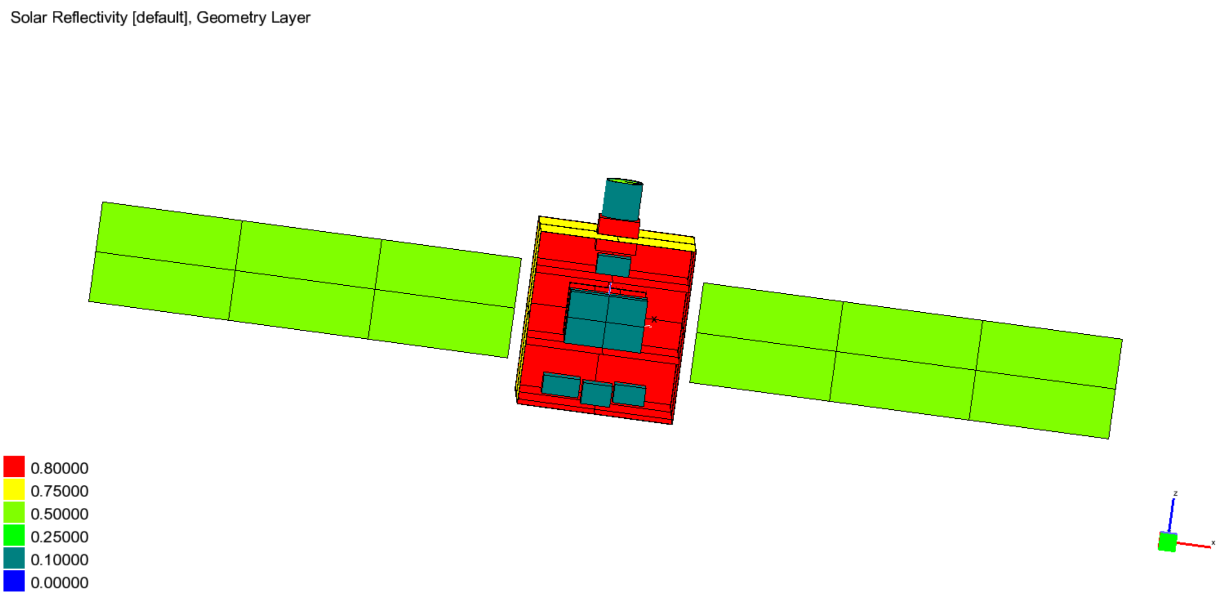


(b) View from the opposite side.

Figure 5.5: Solar absorptivity distribution over the final spacecraft model, viewed from opposite sides.



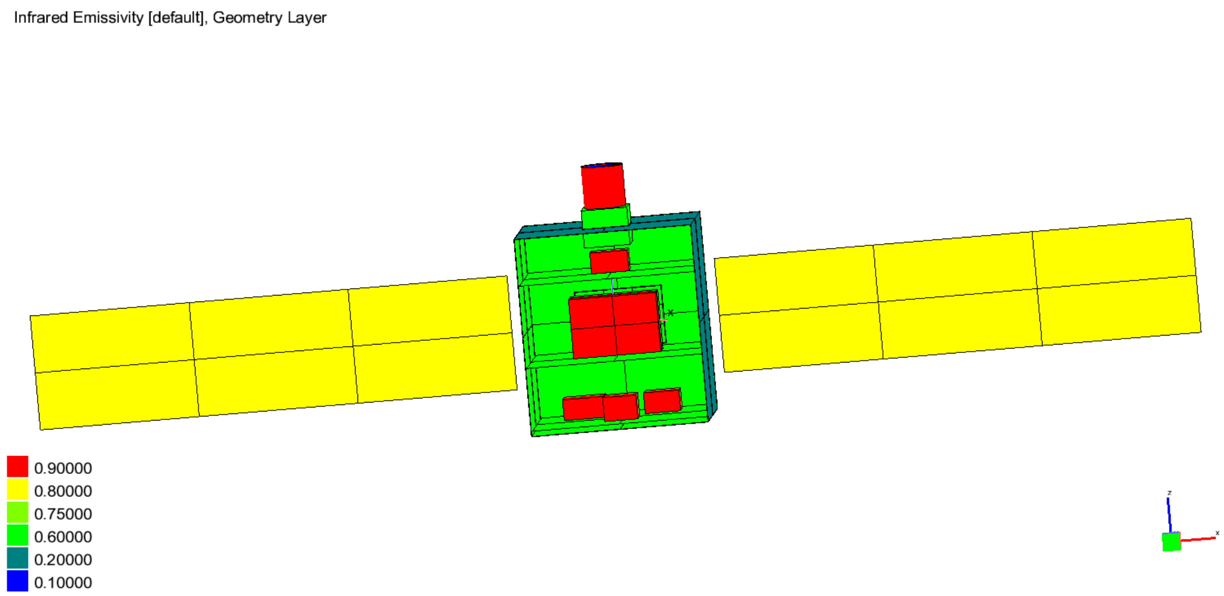
(a) View from one side.



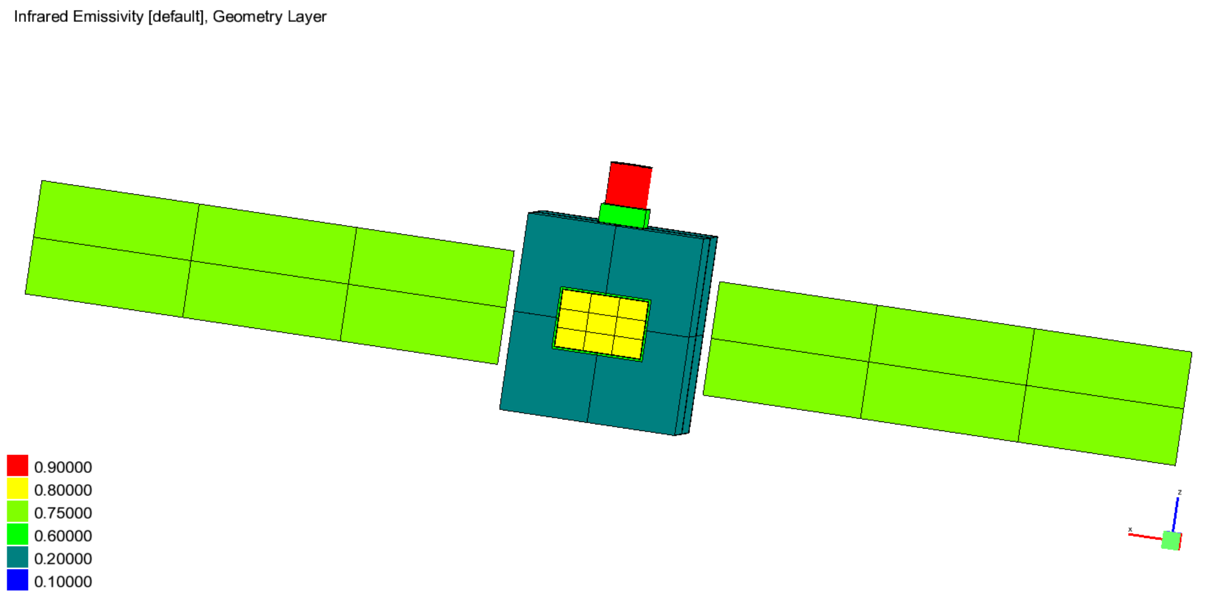
(b) View from the opposite side.

Figure 5.6: Solar reflectivity distribution over the final spacecraft model, viewed from opposite sides.

The infrared emissivity and reflectivity distributions are shown in Figures 5.7 and 5.8. High emissivity is assigned to the black-painted surfaces and the radiator, promoting radiative heat exchange. Lower values are used for the internal MLI and the camera lens, which also includes infrared transmissivity. These assignments determine the radiative exchange between internal elements and the rejection of heat from the external surfaces.

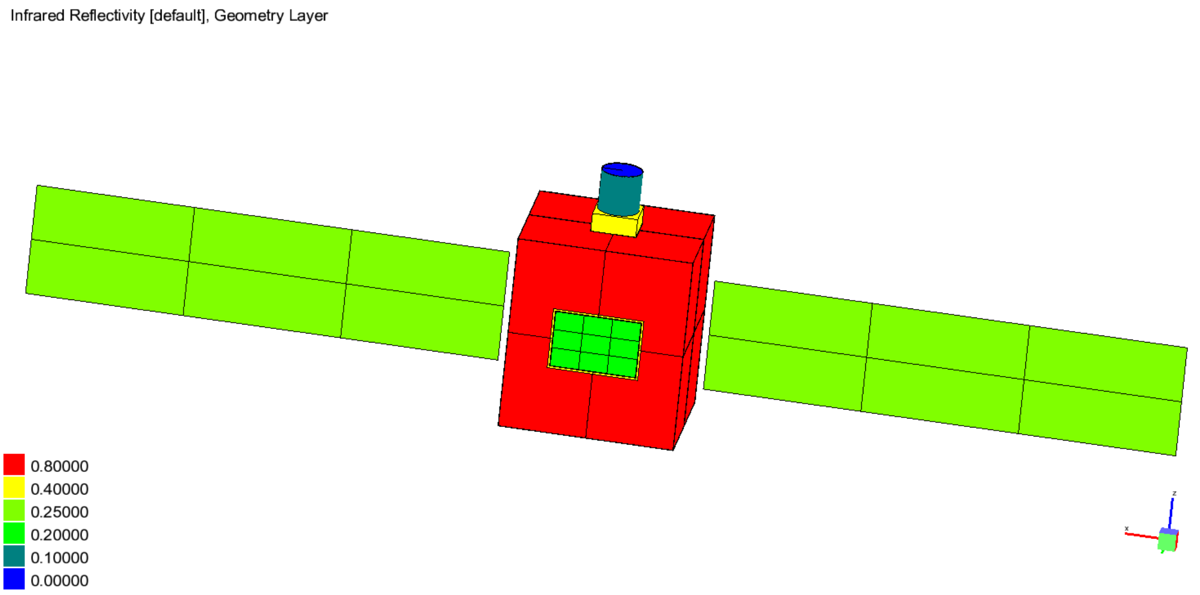


(a) View from one side.

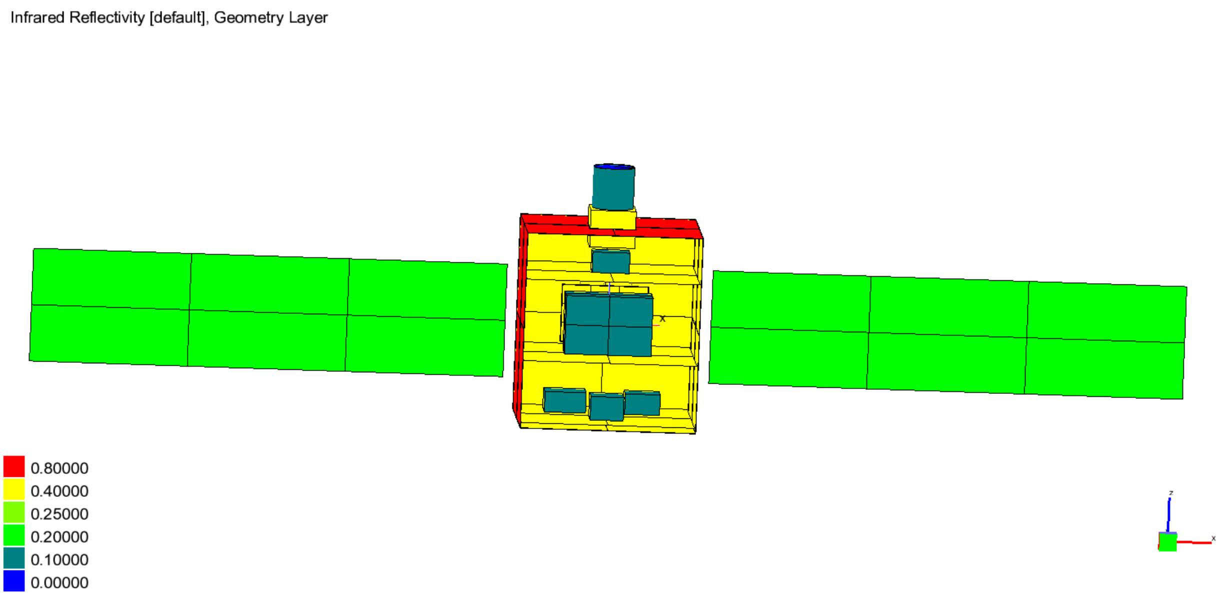


(b) View from the opposite side.

Figure 5.7: Infrared emissivity distribution over the final spacecraft model, viewed from opposite sides.



(a) View from one side.



(b) View from the opposite side.

Figure 5.8: Infrared reflectivity distribution over the final spacecraft model, viewed from opposite sides.

5.4. Nodal Discretisation and Thermal Network Definition

The geometrical model is divided into nodes according to the spatial resolution required for the analysis. Large external elements, including the body panels and solar-panel wings, are represented by several nodes to capture local temperature differences and variations in the absorbed environmental loads. Internal plates and equipment units are also discretised separately to reproduce the main heat-transfer paths through the structure.

Each capacitive node is assigned a material, mass and specific heat capacity. Following this nodal division, the conductive network represents heat transfer through the structure and between the internal equipment and supporting plates. Most conductive paths correspond to physical contacts included in the geometry. When a physical connection is not represented geometrically, an additional conductor is defined. This is the case for the thermal link between the OBC and the radiator, which is included without adding unnecessary geometrical detail.

The radiative network represents exchange between the relevant surfaces and with the deep-space boundary.

Internal heat loads are applied to the nodes representing the dissipative equipment. The values assigned in each transient case are summarised in Table 5.4. The battery load is defined separately for eclipse and non-eclipse intervals.

Table 5.4: Internal heat loads applied in the selected transient cases.

Component	Applied power (W)		
	Nominal	Hot	Cold
Camera electronics	2.00	3.50	0.00
Battery, outside eclipse	1.50	2.06	0.375
Battery, during eclipse	10.00	13.75	2.50
OBC	4.00	5.00	2.00
Communications unit	3.00	4.50	0.50
ADCS	2.00	3.50	0.50

The node definitions and conductive and radiative couplings remain unchanged in all transient cases. Only the environmental conditions and internal heat loads are modified as requires for each case. Consequently, each node and thermal coupling represents the same physical element in every exported result file, allowing direct comparison between cases.

5.5. Orbital Environment and Radiative Cases

The orbital environment defines the external radiative conditions applied to the model. Direct solar radiation, planetary albedo and planetary infrared emission are included as external heat inputs [18]. Energy emitted by the external surfaces towards deep space is also considered. These exchanges are represented through the orbital and radiative models of ESATAN-TMS and provide the boundary conditions used in the transient simulations [8].

Three radiative cases are selected: `rad_nominal`, `rad_hot_operational` and `rad_cold_operational`. The same geometry and surface properties are retained in all three cases, while the environmental parameters and eclipse duration differ between them. For the radiative calculation, each orbital revolution is divided into 24 positions separated by 15°. This common angular sampling allows the incident loads to be evaluated consistently throughout the orbit and included in the transient solution.

Figure 5.9 shows the orbital configuration of the nominal case. The spacecraft positions illustrate the discretisation used for the radiative calculation. The green segment represents the portion outside eclipse, whereas the red segment identifies the eclipse interval.

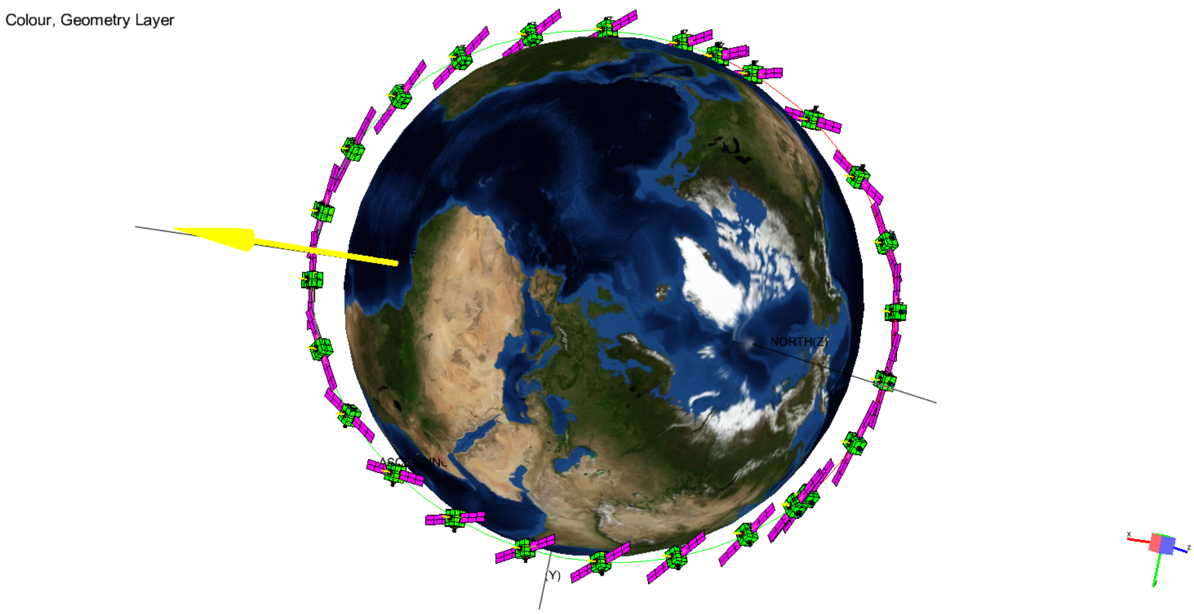


Figure 5.9: Nominal orbital configuration and the 24 positions used for the radiative calculation. The green segment represents the portion outside eclipse, while the red segment indicates the eclipse interval.

The `rad_nominal` case provides the reference environment. `rad_hot_operational` combines a higher external heat input with a shorter eclipse, whereas `rad_cold_operational` combines a lower input with a longer eclipse.

These environments provide clearly differentiated conditions for comparing the hot and cold responses with the nominal case without modifying the physical model. They are selected for the analyses performed in this work and do not cover every condition expected during a flight mission.

5.6. Transient Analysis Cases

5.6.1. Definition of Analysis Cases

Three transient simulations are selected for the final analysis: nominal, hot and cold. The nominal case provides the reference response, while the hot and cold cases represent higher and lower external heat inputs, respectively. Different equipment loads are also assigned according to the operating conditions defined for each simulation. Their purpose is summarised in Table 5.5.

Table 5.5: Transient cases selected from the common thermal model.

Case	**Radiative case**	**Purpose**
Nominal	`rad_nominal`	Reference response used for comparison with the hot and cold cases.
Hot	`rad_hot_operational`	Response under higher external heat input and a shorter eclipse.
Cold	`rad_cold_operational`	Response under lower external heat input and a longer eclipse.

The same geometry, materials, surface properties, node labels, hierarchy and thermal couplings are retained in all three simulations. Only the radiative environment and internal heat loads change between cases. Equivalent nodes and variables therefore retain the same physical meaning, allowing the hot and cold results to be compared directly with the nominal case.

5.6.2. Solver Configuration and Temporal Sampling

The *Solution Control* configuration applied to the three analysis cases is shown in Figure 5.10.

```
C
      TIMEND=6050.2473
      OUTINT=60.502473
      NLOOP=100
      RELXCA=0.01
      DTIMEI=1.0
      CALL SOLCYC('SLCRNC',0.01D0,0.1D0,6050.2473D0,1000,' ','NONE')
C
      CALL SLCRNC
C
```

Figure 5.10: *Solution Control* execution block used to obtain cyclic convergence and calculate the final transient orbital period in ESATAN-TMS.

The same numerical settings are used in all three simulations. The transient solution is calculated using the Crank–Nicolson method. Before the final results are exported, `SOLCYC` repeats the orbital period until the maximum difference between the initial and final nodal temperatures is below 0.01 °C. A maximum of 1000 cycles is allowed. The converged temperature field is then used as the initial condition for the orbital period retained for analysis.

Each exported simulation covers 6050.247 s and contains 101 output instants separated by approximately 60.50 s. This common time basis records the main changes in the applied loads and allows values from different cases to be compared at equivalent points of the orbit.

5.7. Thermal Model Verification

Verification is performed throughout model development using visual and numerical checks. The material and surface-property maps are inspected to identify missing or incorrectly assigned faces. The node hierarchy is compared with the physical arrangement to confirm that the external panels, internal plates and equipment units belong to the correct model groups.

The conductive and radiative networks are reviewed to identify isolated nodes, missing links and unintended heat-transfer paths. Nodal energy balances are examined to detect inconsistent loads or exchanges. Finally, the exported results are checked to confirm that every selected case satisfies the required cyclic-convergence criterion.

These checks confirm the internal consistency of the model but do not validate it against experimental data. No corresponding flight hardware or thermal-vacuum test data are available for comparison. The conclusions are therefore limited to the behaviour predicted by the model under the defined environmental and operating conditions.

5.8. Simulation Results Export in TMD Format

After each simulation is completed, the result set obtained for each case is exported from ESATAN-TMS as a separate `.TMD` file. Each file stores the time vector and the nodal quantities selected for post-processing, including temperatures, applied loads, absorbed environmental loads, and conductive and radiative heat flows. Node numbers and labels are preserved during export, allowing each variable to be associated with the corresponding group in the external hierarchy spreadsheet.

The structural and temporal dimensions shared by the three result files are summarised in Table 5.6. The nominal, hot and cold files retain the same node and conductor structure and cover a common time interval.

Together, the three files form the dataset used for the transient and multi-case analyses. Their common structure allows equivalent variables to be represented at the same orbital instants and compared directly with respect to the nominal case.

Table 5.6: Structural and temporal characteristics of the exported `.TMD` files.

Total thermal nodes per file	284
Capacitive nodes	282
Boundary nodes	1
Inactive nodes	1
Linear conductors, G^L	597
Radiative conductors, G^R	3292
Time interval	6050.247 s
Output instants per file	101
Approximate output time step	60.502 s

The 101 output instants correspond to the temporal sampling of the transient solution and should not be confused with the 24 orbital positions used in the radiative calculation. For each nodal variable, one file therefore contains $284 \times 101 = 28\,684$ values before the conductor results and remaining nodal quantities are considered.

The absorbed solar loads calculated for the solar panels are also retained in the exported data. These values are used as inputs for the electrical-generation and battery calculations. This processing is performed independently and does not modify the ESATAN-TMS solution.

The three `.TMD` files and the hierarchy spreadsheet form the input dataset used by the Python application to read, organise and compare the results.

6. Development of the Multi-Case Post-Processing Tool

This chapter describes the transformation of the original HELIOS post-processing application into a tool capable of loading, inspecting and comparing several transient cases associated with a common ESATAN-TMS model. The development retains the hierarchy-based selection and instantaneous heat-flow views of the starting code, while it reorganises file loading, model storage, interface modules and output management around a shared collection of cases.

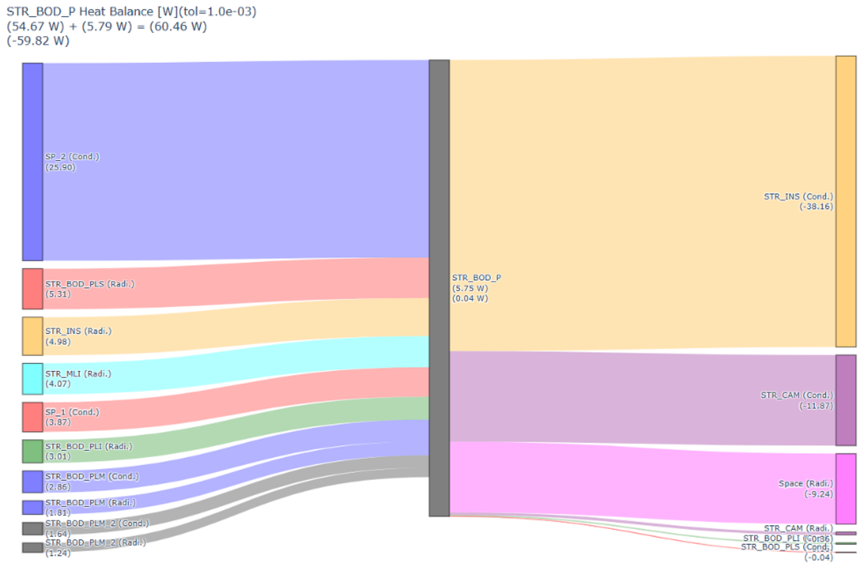
The application reads each `.TMD` file independently and links its numerical results to the hierarchy defined in an external Excel spreadsheet. This structure allows the same physical selection, output instant and representation criteria to be applied to equivalent elements in every loaded case. Direct overlays and reference-minus-comparison results can therefore be generated without merging or modifying the source files.

ESATAN-TMS continues to perform the numerical solution. Python does not repeat the nodal calculation or alter the imported results. Its role is to organise the available data and provide the selection, visualisation, persistence and export functions required for post-processing. The transient and electrical-analysis modules use this common architecture and are developed in the following chapters.

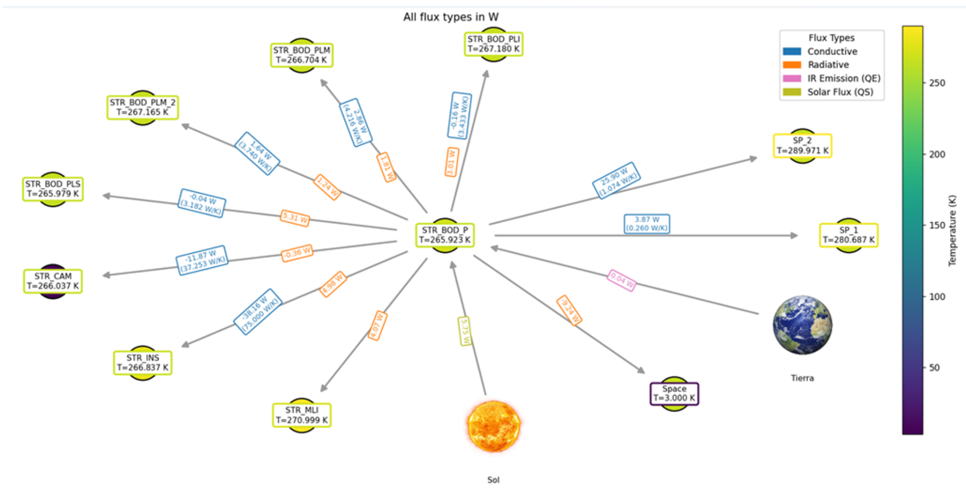
6.1. Baseline and Post-Processing Code Evolution

The work starts from the Python-based application developed for the steady-state analysis of the HELIOS model. Its workflow combines a solved `.TMD` file with an Excel spreadsheet that assigns numerical node identifiers to physical components and subsystems. The application then uses this hierarchy to filter results and generate temperatures, group attributes, conductor tables and heat-flow representations [1].

The result reader, hierarchy-based filters and main heat-flow views provide a valid basis for the present development. Figure 6.1 shows two representative outputs generated for HELIOS. These functions are retained because their physical interpretation remains valid when they are applied to one selected instant of a transient solution.



(a) Sankey representation of the HELIOS heat balance.



(b) NX diagram of HELIOS including all heat-flow types.

Figure 6.1: Representative heat-flow visualisations generated for the HELIOS steady-state case with the original application. Extracted from [1].

The original interface mainly organises the analysis around one result file. Although some routines can read more than one output instant when the source contains them, transient handling does not form part of a coordinated application workflow. Case selection, time selection and output storage also remain independent from one another.

The present model introduces three requirements that affect the complete application rather than a single plotting function. First, several result files must remain available at the same time. Second, every tab must apply the same physical hierarchy and interpret the selected instant consistently.

Third, each saved result must retain enough information to identify its cases, physical elements and representation settings.

The development therefore preserves the useful numerical and graphical routines but replaces the single-file workflow that surrounds them. The final application supports any number of compatible `.TMD` cases within the available input slots, provided that they describe the same model hierarchy and contain the analysis identifier entered by the user. Nominal, hot and cold cases are used throughout this work as representative examples rather than as a fixed limit of the implementation.

6.2. Application Architecture and Data Flow

A modular architecture separates file interpretation, application coordination, shared processing, analysis tabs and local persistence. This organisation keeps the entry point concise and allows the inherited routines to coexist with the transient and electrical modules without concentrating the complete workflow in one script.

Execution starts in `app_thermalviewer/app_interactiva.py`. The Streamlit entry point receives the hierarchy spreadsheet, the internal analysis name and the selected `.TMD` files. It then creates the common case collection, maintains the session state and passes the required data and helper functions to each tab.

`core/thermaldata.py` interprets each result file and provides the nodal, group and conductor information required by the interface. The application keeps the data from every `.TMD` file as an independent case and links all of them to the hierarchy obtained from Excel. The analysis tabs can therefore access several compatible cases during the same session without merging their numerical results or reopening the source files for each operation.

Shared operations are concentrated in `app_helpers.py` and `viewer_utils.py`. They provide hierarchy ordering, descendant searches, redundant-selection removal, unit handling, case-representation controls, numerical subtraction, validation and common output preparation. The modules inside `tabs/` focus on the selections and results specific to each analysis area.

Scene persistence follows a separate branch. `scene_manager.py` manages the SQLite records and files stored in `scene_store/`; `scene_ui.py` provides the saving and history controls; and `restored_scenes.py` returns a stored configuration to the corresponding widgets. Table 6.1 summarises the role of the main groups of files.

Table 6.1: Main software modules and responsibilities in the final application.

Module or directory	Main responsibility
`app_interactiva.py`	Coordinates inputs, loaded cases, Streamlit session state and tab execution.
`core/thermaldata.py`	Reads each result file and provides access to nodes, attributes, time histories and conductor data.
`app_helpers.py`, `viewer_utils.py`	Supply hierarchy, validation, formatting, unit, comparison and shared output functions.
`tabs/`	Contains the model-inspection, heat-flow, transient and electrical analysis interfaces.
`scene_manager.py`, `scene_ui.py`, `restored_scenes.py`	Store, list and restore complete analysis configurations and their generated artifacts.

Figure 6.2 summarises the resulting data flow. The result reader processes the input files as independent cases, the application entry point coordinates their use across the interface, and the shared functions support the different analysis tabs. A separate persistence branch stores generated outputs and returns saved configurations to their source modules.

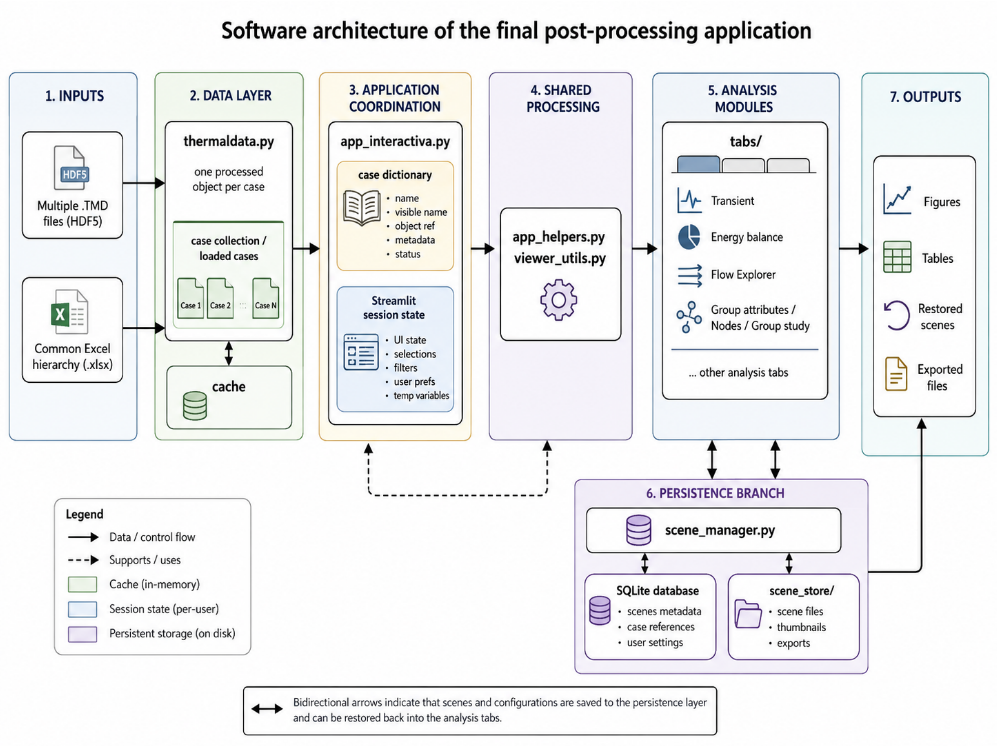


Figure 6.2: Main data flow and software architecture of the current post-processing application.

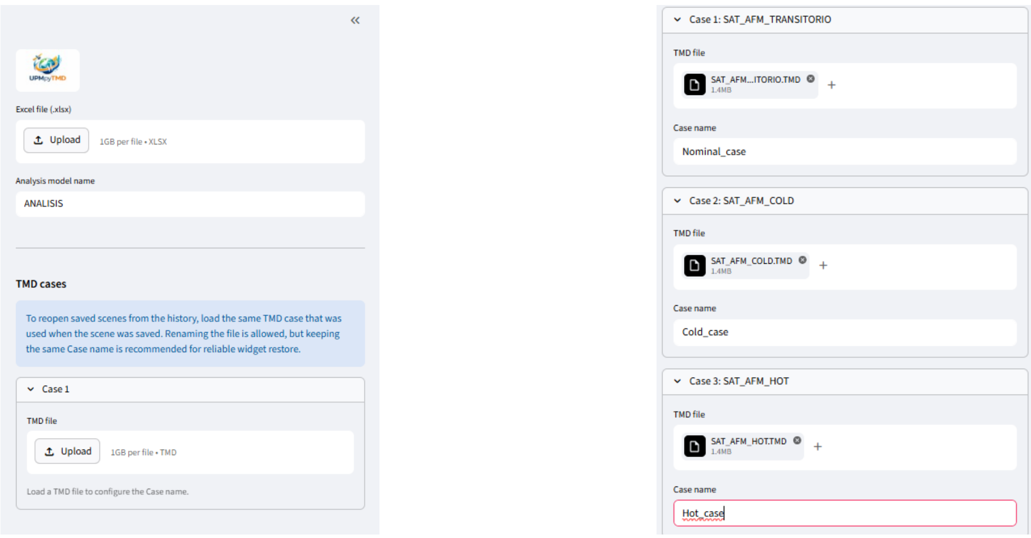
The application keeps imported results separate from values calculated during post-processing. A change in selected groups, units, output time or graphical configuration only affects the current representation. It does not modify the `.TMD` file or the processed object from which the values are read. This separation also allows several tabs to reuse the same cases without introducing inconsistent intermediate copies.

6.3. Multi-Case Loading and Case Organisation

The input stage combines one hierarchy spreadsheet with several transient result files. A common Excel file is sufficient because the cases used in this work retain the same node identifiers, physical groups and parent–child relationships. Environmental and operating conditions change between simulations, but the model organisation remains unchanged.

The interface distinguishes the internal analysis name from the visible case name. The internal name identifies the result structure stored inside each `.TMD` file, whereas the visible name identifies the simulation in the application. Different files may therefore contain the same internal identifier, such as `ANALISIS`, while appearing as *Nominal*, *Hot* or *Cold* in selectors, legends, tables and saved scenes.

Each uploaded file opens an editable case-name field before the processed object is created. When two visible names coincide, the interface adds a suffix so that the keys in the case collection remain unique. Figure 6.3 shows the initial inputs and the subsequent configuration of several files.



(a) Initial hierarchy and result-file inputs. **(b)** Configuration of several files through editable case names.

Figure 6.3: Multi-case loading interface combining a common hierarchy spreadsheet with several `.TMD` result files.

The loader reads every file separately and preserves its results as an independent case. The interface can then request the same variable or physical selection from one or more cases without combining their original numerical data.

The Excel hierarchy performs a second function beyond displaying readable names. It relates each physical group to its node range and parent–child position within the model. Since all loaded cases share this structure, the application can apply the same physical selection to each of them.

For example, selecting `BODY` does not require the user to enter the corresponding node numbers separately for the nominal, hot and cold cases. The hierarchy resolves the group once, and each case supplies the values of those nodes at the requested time. The resulting tables or curves therefore refer to the same physical part of the model before any overlay or subtraction is performed.

The hierarchy also controls ordering and prevents double counting. Selectors follow the physical parent–child sequence defined in Excel rather than a purely alphabetical list. When a parent group and one of its descendants are selected together, the application removes the descendant because the parent already contains its nodes. A visible warning identifies the corrected selection. This check is particularly important for heat-flow and aggregated load results, where repeated nodes would otherwise alter the numerical total.

6.4. Final Application Interface

The final interface uses one layout throughout the complete workflow. The sidebar contains the shared inputs, while the main area contains independent tabs. Since every module receives the same case dictionary and hierarchy, the user can move between model inspection, transient processing and electrical analysis without reloading the files.

Figure 6.4 shows the resulting interface after loading the common hierarchy and three transient cases. The sidebar retains the shared model inputs, whereas the main area concentrates the case, hierarchy, time and representation controls associated with the active analysis tab.

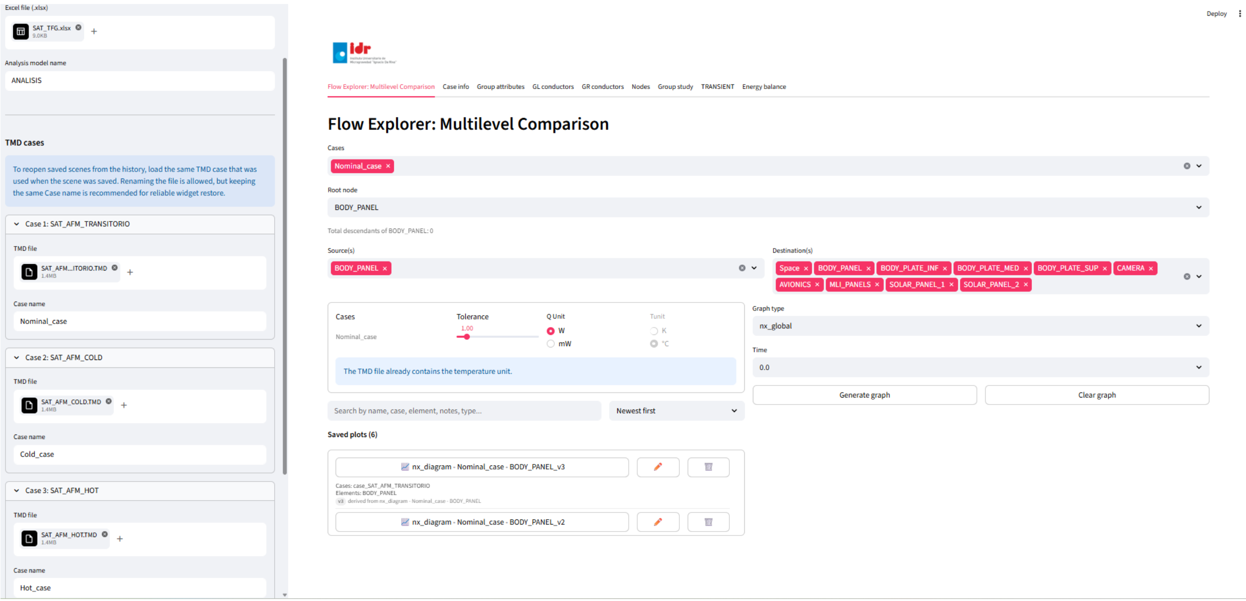
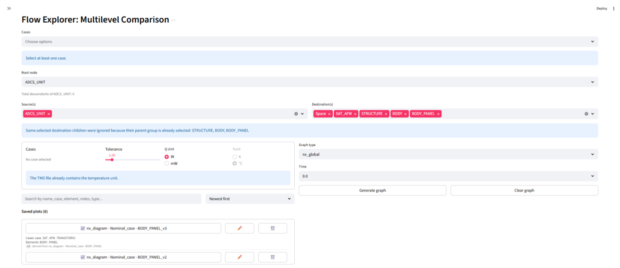


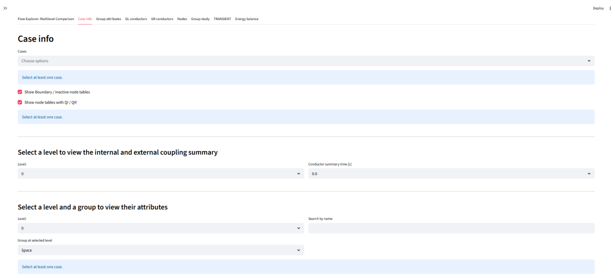
Figure 6.4: General organisation of the final post-processing application after loading three transient cases. The sidebar contains the common hierarchy, the internal analysis name and the result files, while the main area provides access to the model-inspection, transient and electrical-analysis modules.

The first group of tabs examines the imported model and retains the main capabilities of the starting application. `Case info` summarises the available node tables, groups and couplings. `Group attributes`, `Nodes` and `Group study` provide progressively more detailed information about selected parts of the hierarchy. `GL conductors` and `GR conductors` inspect conductive and radiative connections, respectively. `Flow Explorer: Multilevel Comparison` uses the same hierarchical selections to create instantaneous Sankey and network views.

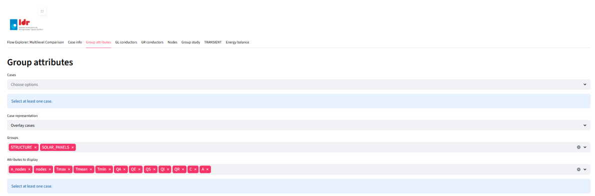
Figures 6.5 and 6.6 show the organisation of these modules after their adaptation to the common multi-case and transient workflow. The screens illustrate the consistent case selector, hierarchy-based controls, time inputs and history areas used across the interface.



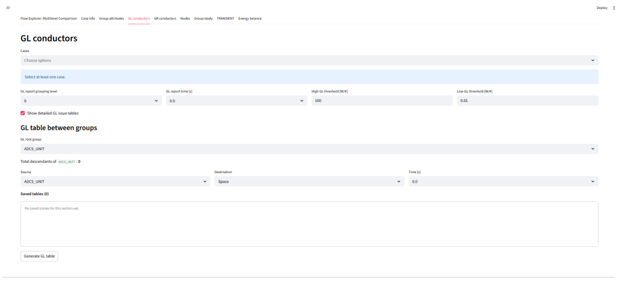
(a) Flow Explorer.



(b) Case information.

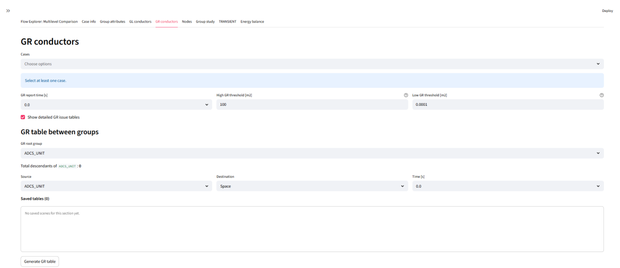


(c) Group attributes.

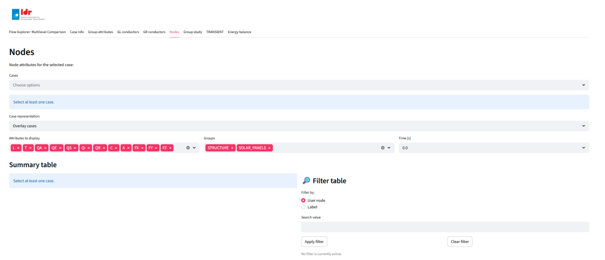


(d) Linear conductors.

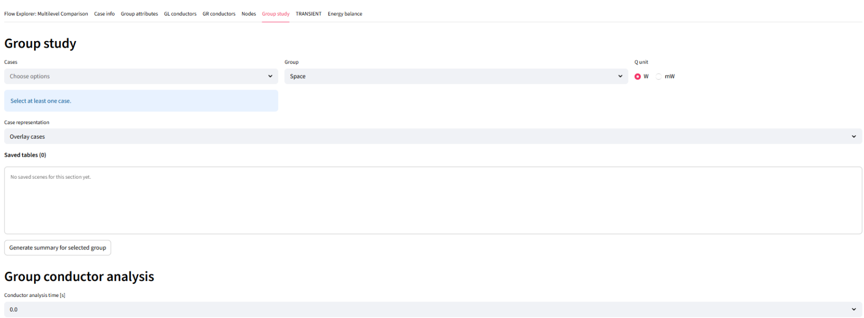
Figure 6.5: General inspection modules adapted to the common transient and multi-case interface.



(a) Radiative conductors.



(b) Nodal information.



(c) Group study.

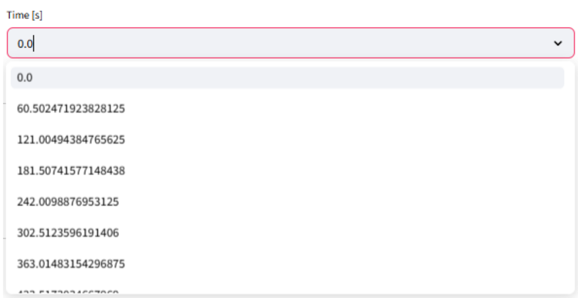
Figure 6.6: Conductor, nodal and group-study modules after integration into the final application.

The controls adapt to the information available in the selected cases. Temperature units already defined in the result file remain fixed, while compatible heat-flow and display units can be changed where appropriate. Validation messages identify empty case selections, unsupported combinations or missing data before a figure or table is generated. These checks provide a uniform response across modules that originally followed separate interaction patterns.

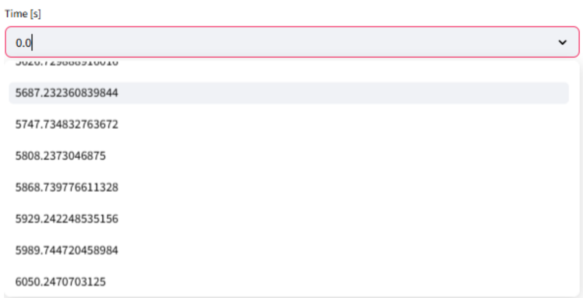
6.4.1. Time Selection

The inherited modules represent the model at one instant, even when their source is a complete transient solution. Each compatible area therefore includes a time selector when the selected result contains more than one output step. The tab obtains the available instants from the transient data and applies the chosen value to attributes, nodal tables, conductor summaries or instantaneous-flow diagrams.

This extension preserves the interpretation of the original steady-state views. Instead of replacing them with complete histories, it allows the user to inspect any stored stage of the transient solution using the same physical and graphical tools. Figure 6.7 shows the selector at the beginning and end of one available time vector.



(a) Initial output instants.



(b) Final output instants.

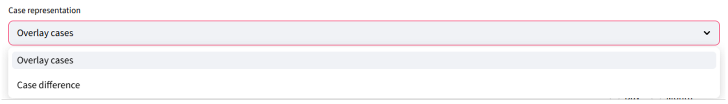
Figure 6.7: Output-time selector used by the instantaneous inspection modules when transient data are available.

When a module compares several cases, it evaluates each selected object at the requested instant available to that analysis. The chosen time is also added to the scene recipe so that restoring a saved result returns the selector to the same position.

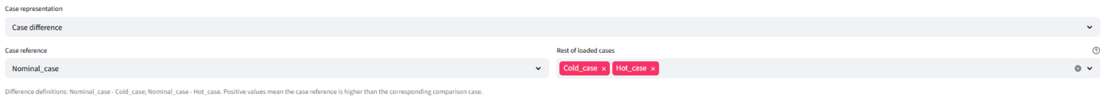
6.4.2. Multi-Case Representation Options

The common case selector allows each compatible tab to work with one or more loaded simulations. In direct representation mode, the module applies the same physical and temporal configuration to all selected cases. Curves can be superimposed when a shared graph supports clear comparison; tables and more complex network outputs can instead remain separated by case.

An additional *Case difference* mode isolates the numerical change between simulations. The user selects one reference case and one or more comparison cases. The application includes the reference and all selected comparisons in the calculation, verifies that the requested variables are compatible and produces one difference result for each pair. Figure 6.8 shows the representation mode and the subsequent reference/comparison controls.



(a) Selection of the case-difference representation mode.



(b) Definition of the reference and comparison cases.

Figure 6.8: Configuration of the case-difference representation.

For any compatible quantity x, the adopted convention is

$$\Delta x_{R-C} = x_R - x_C, \tag{6.1}$$

where R denotes the reference case and C denotes one comparison case. A positive value means that the reference result is greater, whereas a negative value means that the comparison result is greater. The application retains the same order in legends, table headings, captions and saved-scene metadata.

Direct representation and case difference answer different questions. An overlay preserves the absolute response of every case and makes common trends visible. The difference removes the shared level and highlights the effect of the modified environmental or operating condition. Both modes use the same hierarchy, units and time settings, which avoids repeating the complete input configuration when the user changes the form of comparison.

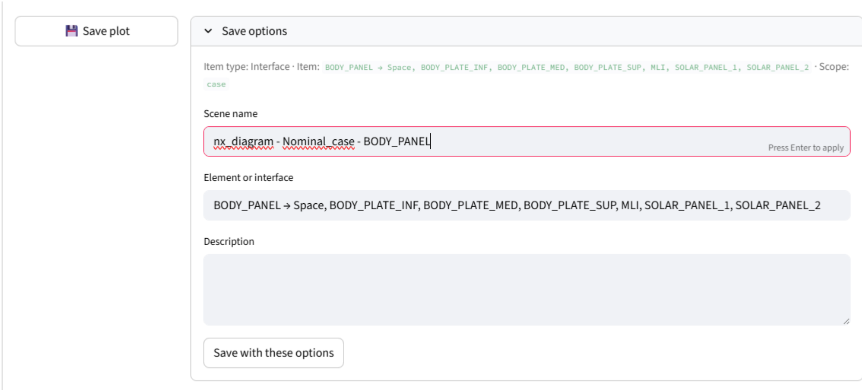
The two final tabs contain the main analysis extensions developed in this work. `TRANSIENT` processes complete histories of temperatures, loads, interfaces and heat flows. `Energy balance` derives solar generation, electrical demand and battery behaviour from the selected cases. Their specific calculations and outputs are presented separately in the following chapters.

6.5. Saving, Restoring and Exporting Analyses

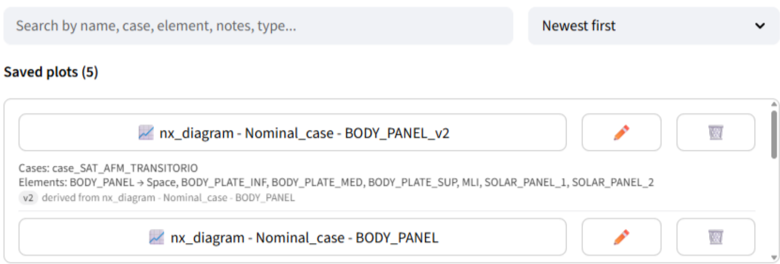
The combination of cases, physical groups, variables, time instants and graphical settings can generate many valid outputs. An exported figure or table preserves the final result but not the complete configuration required to reconstruct it. The application therefore adds a local scene history to the analysis areas that produce reusable outputs.

Once a figure or table has been generated, the corresponding control opens a saving form adapted to the output type. The command appears as *Save plot*, *Save table* or a combined option when both artifacts belong to the same analysis. The user assigns a descriptive name and may add a short explanation. The interface also records the source tab, selected cases and physical element or interface inferred from the active controls.

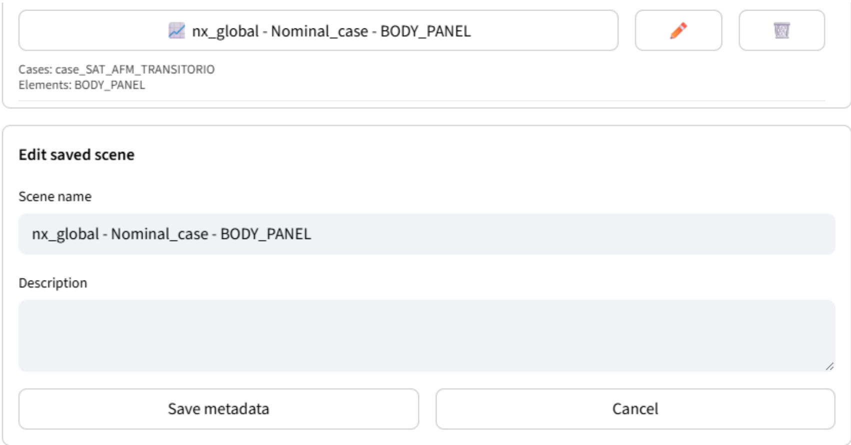
Figure 6.9 shows the complete saving sequence. The first view contains the information entered before storage. The second shows an original scene and a later version derived from it. The third illustrates how the name and description can be edited without changing the stored numerical result.



(a) Definition of the scene name, involved elements and optional description before saving the output.



(b) Saved-scene history containing an original analysis and a subsequently derived version. Selecting a scene name restores its configuration, while the adjacent controls allow its metadata to be edited or the entry to be deleted.



(c) Editing of the name and description associated with an existing scene.

Figure 6.9: Sequence used to save, restore and manage an analysis output within the section history.

The persistence layer separates the scene records from their generated artifacts. The SQLite database stores the saved analyses, their associated cases and the references required to locate their outputs. The directory `scene_store/` contains the generated files and the metadata needed to restore each configuration.

Every scene includes a recipe that records the widget values needed to recreate the analysis. Depending on the source tab, the recipe can contain case names, hierarchy selections, variables, units, time values, comparison mode and plot options.

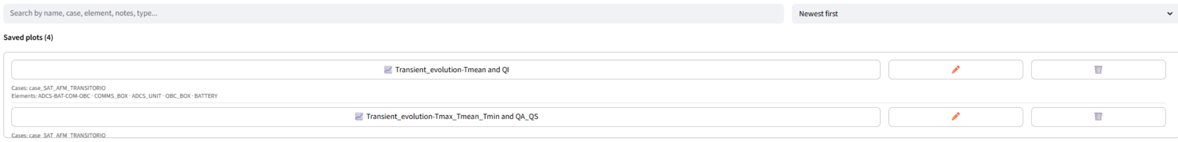
The scene name displayed in each history entry acts as the restoration control. Selecting it loads the stored recipe into the source tab and recovers the associated cases, hierarchy selections, units, time values and representation settings. The adjacent pencil and bin controls modify the scene name and description or delete the entry, respectively. The same result files must be loaded, and retaining the original visible case names improves the reliability of widget restoration.

The storage format follows the output type. Plotly figures are saved as JSON and HTML, data tables as CSV and Excel files, and Matplotlib figures as PNG and SVG. Each scene also receives a `scene.json` file that contains its name, description, scope, cases and reconstruction recipe. This combination supports both interactive recovery inside the application and direct access to conventional engineering file formats.

Restoring a scene does not overwrite the original entry. If the user changes its recovered configuration and saves the new result, the application creates a separate scene and adds the source-scene information to the new recipe. The history can therefore show the derived relationship while preserving the original analysis.

Search and ordering controls act on the complete history. Entries can be located through their name, case, physical element, description or output type, and each item can be renamed or deleted from the same area. The history mechanism is common to the application, although each tab stores the artifacts that correspond to its own analysis.

Figure 6.10 shows the same system applied to two transient result areas. One history contains parameter plots, whereas the other combines heat-flow figures with associated tables.



(a) History containing different parameters generated from transient results.

(b) History containing transient-flow figures and associated tabular results.

Figure 6.10: Examples of saved-scene management in different transient result areas.

Scene storage and file export fulfil complementary purposes. The history preserves the configuration and relationships required to continue an analysis inside the application. Download controls create external copies of figures or tables for reports and engineering documentation. Together, both functions connect interactive exploration with reproducible post-processing and final result presentation.

The resulting chapter-level workflow can be summarised as follows: the application loads several compatible cases, associates them with one physical hierarchy, applies common selections and time settings, generates direct or difference outputs, and stores the complete configuration whenever the analysis must be recovered. This architecture provides the common foundation for the transient and electrical modules developed next.

7. Transient Thermal Analysis and Multi-Case Comparison

7.1. Transient Data Processing

7.2. Temperature and Thermal-Load Evolution

7.3. Interface Temperature-Difference Analysis

7.3.1. Interfaces between Model Components

7.3.2. Variation over Selected Time Intervals

7.4. Thermal Summary Tables

7.5. Conductive and Radiative Heat-Flow Evolution

7.6. Thermal Energy and Summary Tables

8. Electrical Power Balance and Battery Model

This chapter presents the electrical power-balance methodology implemented in the post-processing tool. Thermal results from the solar panels and the battery are used to estimate electrical generation, power demand, and battery performance throughout the simulated orbital cycle.

8.1. Method Overview and Thermal Inputs

a

8.2. Solar Power Conversion

8.2.1. Input Properties and Conversion Efficiency

8.2.2. Incident and Electrical Power Calculation

8.3. Electrical Loads and Net Power Balance

8.4. Battery Model and Operating Strategy

8.4.1. Pack Configuration and Electrical Parameters

8.4.2. Voltage, Internal Resistance and State of Charge

8.4.3. Charge and Discharge Control

8.4.4. Temperature Effects on Battery Performance

8.5. Calculation Procedure and Output Variables

9. Results

9.1. Verification of the Post-Processing Tool

9.2. Transient Thermal Results

9.2.1. Temperature Evolution and Case Comparison

9.2.2. Effect of the Battery Heater

9.2.3. Interfaces, Heat Flows and Thermal Energy

9.3. Electrical Power and Battery Results

9.3.1. Solar Generation and Load Balance

9.3.2. Battery Performance

10. Conclusions and Future Work

10.1. Conclusions

10.2. Future Work

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